01 Introduction

Stocklane is a workshop stock-management platform that connects a mobile-friendly React frontend to Spring Boot and FastAPI microservices running on Amazon EKS. It supports receiving parts, giving parts to workshop jobs, catalogue search, barcode and QR recognition, photo identification, reviewable text-extraction suggestions, stock counting, movement history and owner reporting.

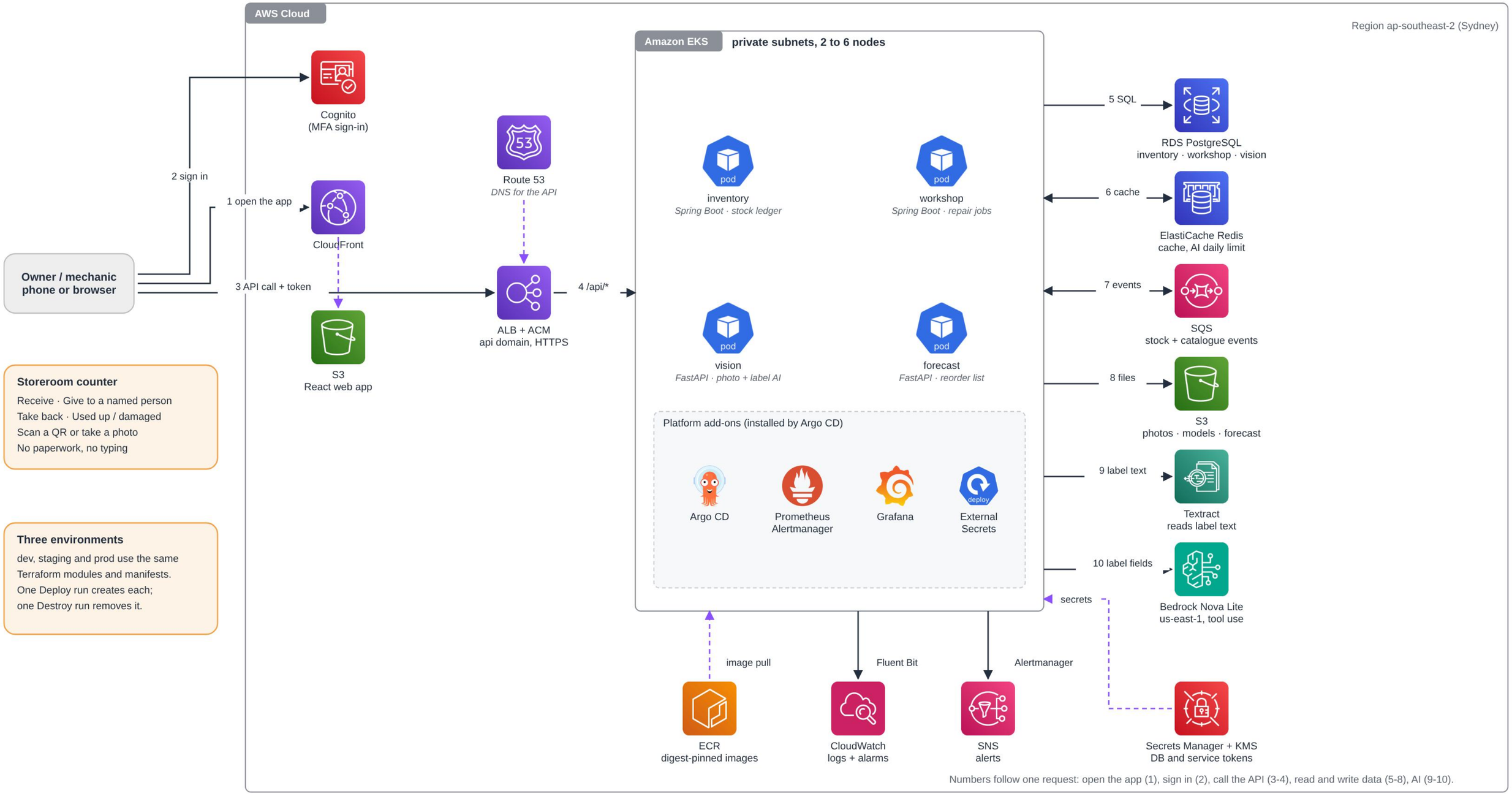
- The platform is designed around a simple rule: AI suggests, a person approves, and the system records the decision.
- React provides the operator experience; Spring Boot services handle core inventory and workshop transactions; FastAPI services handle vision and forecast inference.
- AWS managed services provide identity, networking, data, messaging, storage, security, model operations and observability.

Zoomed-out Stocklane platform

The zoomed-out architecture shows the complete platform. The React web client is the operator entry point. Spring Boot microservices provide inventory, catalogue and workshop APIs, while FastAPI services provide photo matching and demand forecasting. EKS runs these services, and AWS data, AI and operations services support them.

- Frontend: React web application for mobile and desktop workflows.
- Backend: Spring Boot inventory/workshop services plus FastAPI vision/forecast services.
- Platform: Amazon EKS, PostgreSQL, Redis, S3, SQS, Cognito, Textract, Bedrock, SageMaker, Prometheus and Grafana.

Stocklane: workshop stock management on AWS (final architecture)

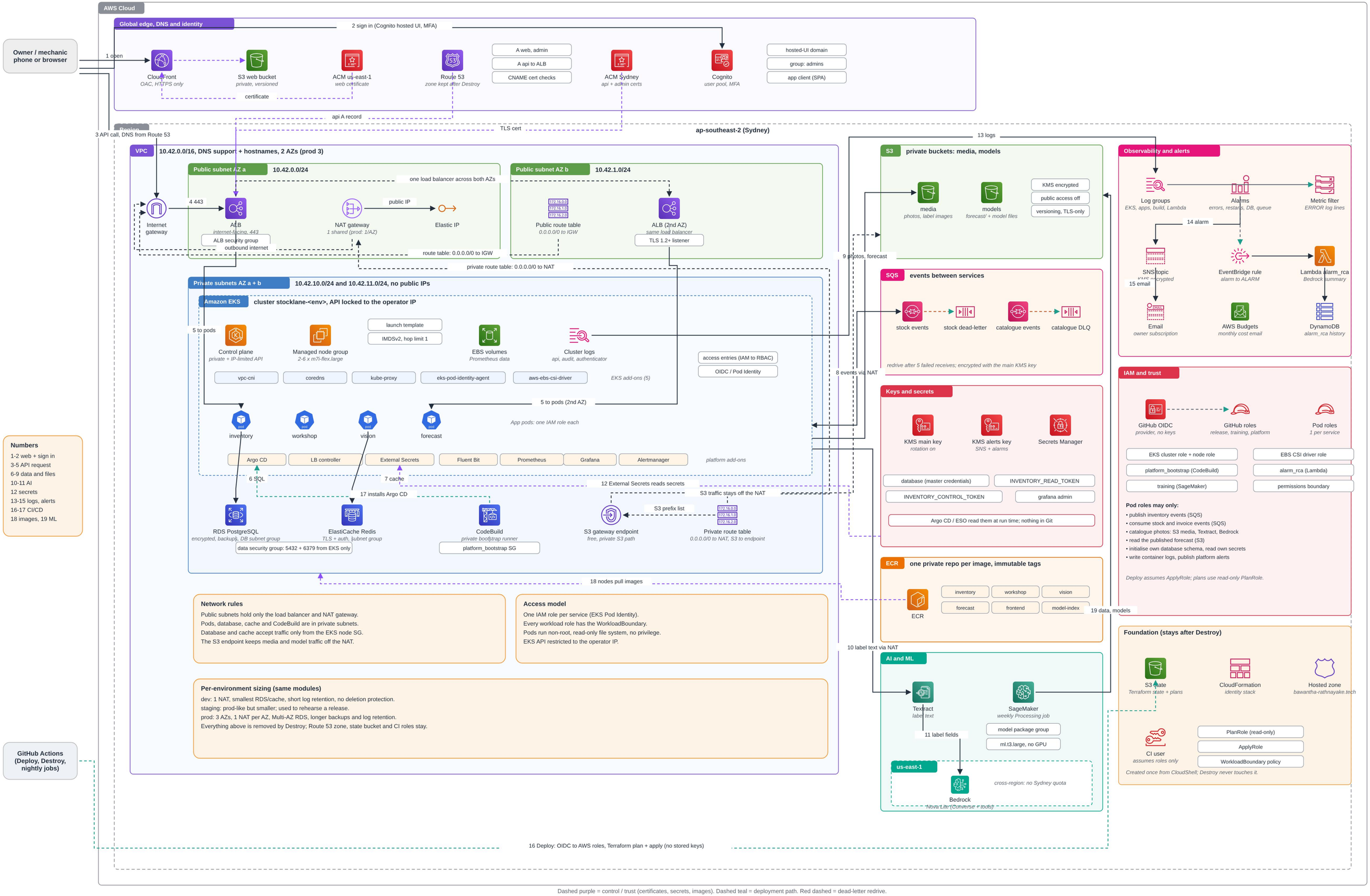


Overall infrastructure and services used

This diagram is the infrastructure index. Public edge and DNS lead to a VPC with public and private subnets. EKS hosts the microservices, while managed AWS services provide persistence, asynchronous processing, identity, AI/ML and operational control.

- Edge and identity: Route 53, Application Load Balancer and Amazon Cognito.
- Runtime and data: EKS, Aurora/RDS PostgreSQL, Redis, EBS, S3 and SQS.
- Security and operations: IAM, KMS, Secrets Manager, CloudWatch, EventBridge, Lambda and SNS.
- AI and ML: Textract, Bedrock and SageMaker processing jobs for label reading, suggestions and model workflows.

Stocklane: complete AWS infrastructure (everything Terraform creates, per environment)



User flow and backend flow

A user signs in with Cognito, opens the React application and chooses a simple stock action. The request travels through the AWS edge into EKS, where Spring Boot services validate the transaction and persist the movement. FastAPI services are called for photo or forecast tasks. S3 stores images and model artefacts; SQS separates asynchronous work; dashboards and alerts expose the result.

- 1. Authenticate: Cognito issues the identity used by the React client and APIs.
- 2. Operate: React sends Receive, Give, search, count or catalogue requests to backend services.
- 3. Process: Spring Boot owns transactional stock logic; FastAPI owns ML inference.
- 4. Assist: Textract/Bedrock produce suggestions; the operator manually approves or corrects them.
- 5. Record and observe: PostgreSQL, S3, SQS, Redis, Prometheus, Grafana and CloudWatch support the workflow.

Stocklane · AWS infrastructure (dev environment)

Everything Terraform creates in one environment · 2 Availability Zones · dashed grey items exist only in prod · numbers 1–10 follow the legend at the bottom

Owner / mechanic
phone or browser

Operator
kubectl / console
allowed IP only

GitHub Actions
Deploy · Destroy
nightly forecast
weekly training
OIDC, no stored keys

AWS Cloud · Region ap-southeast-2 (Sydney)

Global edge, DNS and identity

CloudFront (1)
HTTPS only
Origin Access Control

OAC

S3 web bucket
React build
private · versioned

Cognito user pool (2)
MFA · groups OWNER, MECHANIC
hosted-UI domain · SPA client

ACM · us-east-1
certificate for
CloudFront

ACM · Sydney
certificates for api
and admin → ALB

Route 53
web, admin → CloudFront
api → ALB
zone kept after Destroy

Keys and secrets

Secrets Manager
database
service tokens
Grafana admin

KMS main key
RDS · S3 · SQS
Redis · secrets
EKS secrets

KMS alerts key
SNS topic
alarm mails

Secrets Manager → External Secrets → pods

S3 · private buckets

media bucket
catalogue photos
label images
reference photos

models bucket
model bundles
training datasets
forecast file

KMS · TLS-only · public access off

IAM and trust

GitHub OIDC
identity provider
no stored keys

GitHub roles
release · training
platform

Pod Identity roles (one per service):
inventory · workshop · vision · forecast
db-bootstrap · external-secrets
LB controller · fluent-bit · alertmanager
EBS CSI

Also: EKS cluster role · node role ·
alarm_rca Lambda role · SageMaker role ·
CodeBuild role · all pod roles share the
WorkloadBoundary permissions boundary

SQS · events

stock events
inventory → workshop

stock DLQ
after 5 failures

catalogue events
inventory → vision

catalogue DLQ
after 5 failures

KMS-encrypted

CI/CD and ML

SageMaker
weekly Processing job
ml.t3.large, no GPU

Model registry
package group
versions + metrics

Automation bucket
deployment metadata
versioned · lifecycle

CodeBuild project
platform bootstrap
own log group

AI · via the NAT gateway

Textract
label OCR (Sydney)
always runs first

us-east-1 · Bedrock

Bedrock
Nova Lite

Converse + tool use
fills missing label fields
max 50 uncached calls/day
no Sydney quota

a person always confirms the draft

Foundation (stays after Destroy)

CloudFormation
CI roles +
state bucket

Terraform state
saved plans
hash-checked

Hosted zone
bawantha-rathnayake.tech

CI roles
Plan (read-only)
Apply · boundary

ECR · images

ECR
immutable
tags

inventory · workshop · vision · forecast · operations · ml-training

worker nodes pull images by digest

VPC 10.42.0.0/16 · 2 Availability Zones · DNS hostnames on

Internet gateway (3)
attached to VPC

egress

NAT gateway
1 shared in dev
+ Elastic IP

HTTPS 443

ONE Application Load Balancer
a node in each AZ

ALB node · AZ A
HTTPS 443 · TLS 1.2+
ACM certificate

ALB node · AZ B
same LB, same DNS name
targets: pod IPs

NAT gateway · AZ B
prod only
(1 NAT per AZ)

Availability Zone A

Public subnet 10.42.0.0/24

Public route table: 0.0.0.0/0 → Internet gateway

Private subnet 10.42.10.0/24

Availability Zone B

Public subnet 10.42.1.0/24

same public route table

Private subnet 10.42.11.0/24

Amazon EKS cluster

stocklane-dev · managed control plane
API endpoint limited to the operator IP

Access entries (IAM → Kubernetes RBAC)

OIDC provider · Pod Identity agent

Managed node · AZ A
m7i-flex.large · EBS gp3
IMDSv2 required (hop limit 1)

Managed node · AZ B
m7i-flex.large · EBS gp3
IMDSv2 required (hop limit 1)

Ingress · /api/inventory · vision · forecast · workshop

Namespace stocklane-dev · Deployments (ClusterIP, 1 replica in dev)

inventory :8081	workshop :8082	vision :8083	forecast :8084
Spring Boot stock ledger	Spring Boot jobs (UI unused)	FastAPI photos + labels	FastAPI reorder forecast

Platform add-ons

Argo CD (GitOps)	AWS LB Controller
External Secrets	cert-manager
Fluent Bit (logs)	Prometheus Operator
metrics-server	CoreDNS
kube-proxy	VPC CNI
EBS CSI driver	NetworkPolicies

Namespace monitoring

Prometheus	Grafana
Alertmanager	kube-state-metrics

SQL 5432 (4)

Redis 6379

EKS cluster security group

RDS PostgreSQL 17
db.t4g.micro · single-AZ
KMS · 1-day backups
schemas: inventory, workshop, vision

ElastiCache Redis 7.1
primary · AZ A
TLS + auth · KMS

Redis replica
AZ B · automatic
failover on

RDS standby
Multi-AZ, prod only

Data security group: 5432 / 6379 only from the EKS cluster security group

DB subnet group + ElastiCache subnet group span both AZs

CodeBuild runner
in-VPC, egress 443 only
installs Argo CD

EBS gp3 volumes
Prometheus + Grafana
persistent volumes

S3 gateway endpoint
free private path to S3

Private route table A: 0.0.0.0/0 → NAT gateway · S3 prefix list → S3 endpoint

Private route table B: 0.0.0.0/0 → NAT in AZ A (dev) · S3 prefix list → S3 endpoint

(9)

Observability and alerts

CloudWatch Logs
EKS control plane · apps
alarm_rca · bootstrap

CloudWatch alarms
queue age · dead letters
application errors

SNS operations topic
TLS only · KMS
also fed by Alertmanager

Owner e-mail
SNS subscriber

AWS Budgets
monthly cost mail
$25 in dev

EventBridge rule
alarm state → ALARM

Lambda alarm_rca
evidence + summary
posts to SNS

DynamoDB table
alarm_rca de-dupe
daily cap

Security groups

ALB SG
443 from the internet

EKS cluster SG
control plane and nodes

Data SG
5432, 6379 only from cluster SG

Bootstrap SG
egress 443, allowed into EKS API

Network ACLs: VPC default
Flow logs: not enabled

Access model

One IAM role per service (Pod Identity)
Every role has the WorkloadBoundary
Pods: non-root, read-only file system
Default-deny NetworkPolicies
EKS API restricted to the operator IP
Secrets read at run time, none in Git

Same modules in every environment, only the sizes change

dev	2 AZ · 1 NAT · 2 × m7i-flex.large · RDS db.t4g.micro single-AZ
	Redis 1 replica · short log retention · $25 budget · no deletion protection
staging	2 AZ · 1 NAT · 2–3 nodes · RDS db.t4g.small · 2 replicas
prod	3 AZ · NAT per AZ · 3–6 × m6i.large · Multi-AZ RDS · 3 replicas
	(autoscaling to 6) · deletion protection · longer backups
all	removed by Destroy; Route 53 zone, state bucket, CI roles stay

(8) Operator → EKS API

deploy over OIDC: Terraform plan + apply, image push, Argo CD sync, SageMaker jobs (10)

How to read the numbers

1. Open the app: CloudFront serves the React build from the private S3 web bucket.
2. Sign in: Cognito hosted UI with MFA; the app gets a token (OWNER or MECHANIC).
3. API call: HTTPS → Internet gateway → ALB (a node in each AZ) → ingress → pods.
4. Data: pods use RDS PostgreSQL and the Redis cache.
5. Files: photos, models and the forecast file live in S3, reached over the S3 endpoint.
6. Events: outbox relay → SQS; a message that fails 5 times goes to its dead-letter queue.
7. Label reading: Textract first, then Bedrock in us-east-1, both through the NAT gateway.
8. Operator: kubectl reaches the EKS API only from the allowed IP address.
9. Logs and alerts: Fluent Bit → CloudWatch → alarms → SNS → e-mail, plus a Lambda root-cause note.
10. Deploy: GitHub Actions assumes AWS roles with OIDC, runs Terraform, builds images, drives Argo CD.

02 DevOps

DevOps makes application and infrastructure changes repeatable from commit to running platform.

- GitHub Actions validates code, builds and releases services, and drives deployment automation.
- Terraform defines the AWS environment as versioned infrastructure.
- CloudFormation-backed Terraform state and trust resources provide a controlled bootstrap path.

02 • DEVOPS

GitHub Actions delivery pipeline

The pipeline is the automated path from a GitHub change to a validated release. It connects source control, tests, image or frontend publishing, Terraform and deployment configuration.

github.com/bawantha395/workshop-stock-management-system-mlops/actions/runs/41042988274

bawantha395 / workshop-stock-management-system-mlops

Type / to search

Code | Issues | Pull requests 9 | Agents | Actions | Projects | Wiki | Security and quality | Insights | Settings

← Deploy environment

Deploy environment #16

Re-run all jobs

Summary

All jobs

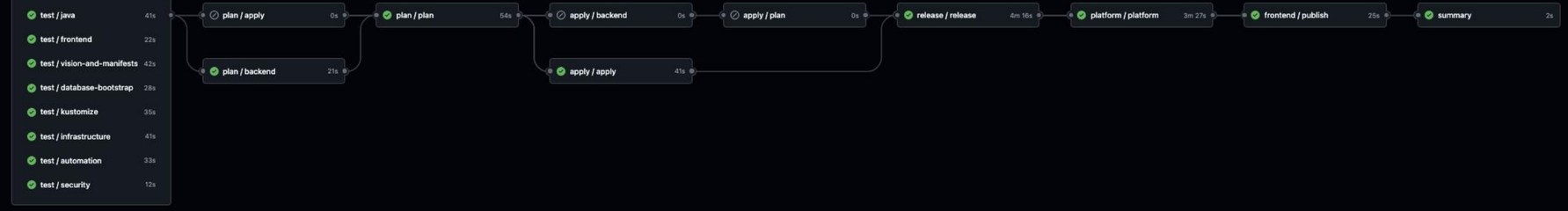
- test
- plan
- apply
- release
- platform
- frontend
- summary

Run details

- Usage
- Workflow file
- Lockfile

Manually triggered 2 hours ago	Status	Total duration	Artifacts
bawantha395 c7d2c1b main	Success	11m 22s	–

deploy.yml
on: workflow_dispatch



Terraform repository structure

The Terraform folder separates reusable infrastructure patterns from environment-specific configuration. This supports reviewable plans, repeatable provisioning and consistent dev/staging/production conventions.

Files

main

Go to file T

- terraform
  - environments
    - dev
      - tests
      - .terraform.lock.hcl
      - main.tf
      - outputs.tf
      - terraform.tfvars
      - variables.tf
      - versions.tf
    - prod
    - staging
  - modules
    - auth
    - cicd
    - data
    - edge
    - eks

workshop-stock-management-system-mlops / infra / terraform /

Add file

bawantha395 validate platform ✓ f35f03c · 18 hours ago History

Name	Last commit message	Last commit date
..		
environments	validate platform	18 hours ago
modules	final changes with ui and prod	18 hours ago

02 • DEVOPS

Terraform backend and trust bootstrap

CloudFormation evidence shows the Terraform backend and GitHub trust resources created successfully. This is the AWS foundation that lets CI use short-lived, controlled access instead of long-lived credentials.

CloudFormation

Stacks
Stack details
Drifts
Stack refactors New
StackSets
Exports

IaC generator
Template studio

Hooks overview
Invocation summary
Hooks

Registry
Public extensions
Activated extensions
Publisher

Spotlight

Feedback

Stacks (2)

Search by stack name

Filter status: Active

View nested

1

Stacks	
stocklane-dev-terraform-backend	2026-10-25 23:53:20 UTC+0530 CREATE_COMPLETE
stocklane-dev-github-trust	2026-10-25 23:33:28 UTC+0530 UPDATE_COMPLETE

stocklane-dev-terraform-backend

Delete stack | Update stack | Stack actions | Create stack

Stack info | Events | Resources | Outputs | Parameters | Template | Changesets

Overview

Stack ID
arn:aws:cloudformation:ap-southeast-2:721274598579:stack/stocklane-dev-terraform-backend/28234c30-d0a1-11f1-843c-0affd105eb3f

Description
Private versioned Terraform state and short-lived saved plans. Managed by backend bootstrap workflow.

Status
CREATE_COMPLETE

Detailed status
-

Status reason
-

Root stack
-

Parent stack
-

Created time
2026-10-25 23:53:20 UTC+0530

Updated time
2026-10-25 23:53:26 UTC+0530

Deleted time
-

Drift status
NOT_CHECKED

Last drift check time
-

Termination protection
Deactivated

IAM role
-

03 GitOps

GitOps keeps the desired Kubernetes state in Git and uses Argo CD to reconcile the EKS cluster. Prometheus and Grafana make the reconciled platform observable.

- Argo CD is the deployment control plane.
- The application tree exposes Kubernetes resources and health.
- Prometheus metrics and Grafana dashboards show API and EKS behaviour.

Argo CD applications and projects

Argo CD shows the applications and platform components managed from Git, including the Stocklane services and monitoring stack.

argo v3.5.6

- Applications
- ApplicationSets
- Settings
- User Info
- Documentation

Application filters

- Favorites Only

SYNC STATUS

- Unknown 0
- Synced 5
- OutOfSync 0

HEALTH STATUS

- Progressing 0
- Suspended 0
- Healthy 5
- Degraded 0
- Missing 0
- Unknown 0

OPERATION STATUS

- Syncing 0
- Sync OK 5
- Sync error 0
- Sync failed 0
- Unknown 0

Applications — APPLICATIONS TILES

+ NEW APP | SYNC APPS | REFRESH APPS | Search applications... | Log out

Sort: name | Items per page: 10

stocklane-dev

Project:	default
Labels:	
Status:	Healthy Synced
Repository:	https://github.com/bawantha395/workshop-stock-man...
Target Revision:	main
Path:	deploy/kubernetes/argocd/applications/overlays/dev
Destination:	in-cluster
Namespace:	argocd
Created At:	10/30/2026 01:04:27 (17 hours ago)
Last Sync:	10/30/2026 01:07:08 (17 hours ago)

SYNC | REFRESH | DELETE

stocklane-monitoring-dev

Project:	stocklane-platform-dev
Labels:	
Status:	Healthy Synced
Repository:	https://github.com/bawantha395/workshop-stock-man...
Target Revision:	main
Path:	deploy/kubernetes/monitoring/overlays/dev
Destination:	in-cluster
Namespace:	monitoring
Created At:	10/30/2026 01:07:04 (17 hours ago)
Last Sync:	10/30/2026 16:25:06 (2 hours ago)

SYNC | REFRESH | DELETE

stocklane-monitoring-operator-dev

Project:	stocklane-platform-dev
Labels:	
Status:	Healthy Synced
Repository:	https://github.com/bawantha395/workshop-stock-man...
Target Revision:	main
Path:	deploy/kubernetes/monitoring-operator/overlays/dev
Destination:	in-cluster
Namespace:	monitoring
Created At:	10/30/2026 01:07:00 (17 hours ago)
Last Sync:	10/30/2026 16:25:20 (2 hours ago)

SYNC | REFRESH | DELETE

stocklane-platform-dev

Project:	stocklane-platform-dev
Labels:	
Status:	Healthy Synced
Repository:	https://github.com/bawantha395/workshop-stock-man...
Target Revision:	main
Path:	deploy/kubernetes/platform/overlays/dev
Destination:	in-cluster
Namespace:	kube-system
Created At:	10/30/2026 01:04:35 (17 hours ago)
Last Sync:	10/30/2026 16:25:06 (2 hours ago)

SYNC | REFRESH | DELETE

stocklane-workshop-dev

Project:	stocklane-app-dev
Labels:	
Status:	Healthy Synced
Repository:	https://github.com/bawantha395/workshop-stock-man...
Target Revision:	main
Path:	deploy/kubernetes/apps/overlays/dev
Destination:	in-cluster
Namespace:	stocklane-dev
Created At:	10/30/2026 01:07:07 (17 hours ago)
Last Sync:	10/30/2026 17:43:39 (38 minutes ago)

SYNC | REFRESH | DELETE

Argo CD resource tree

The resource tree shows how a Git-defined application becomes Kubernetes Deployments, Services, Ingress, autoscalers and supporting policy resources.

argo v3.5.6

- Applications
- ApplicationSets
- Settings
- User Info
- Documentation

Resource filters

NAME

NAME

KINDS

KINDS

SYNC STATUS

- Synced 33
- OutOfSync 0

HEALTH STATUS

- Progressing 0
- Suspended 0
- Healthy 21
- Degraded 0
- Missing 0
- Unknown 0

APP HEALTH
Healthy

SYNC STATUS
Synced to main (647bae8)

Auto sync is enabled.

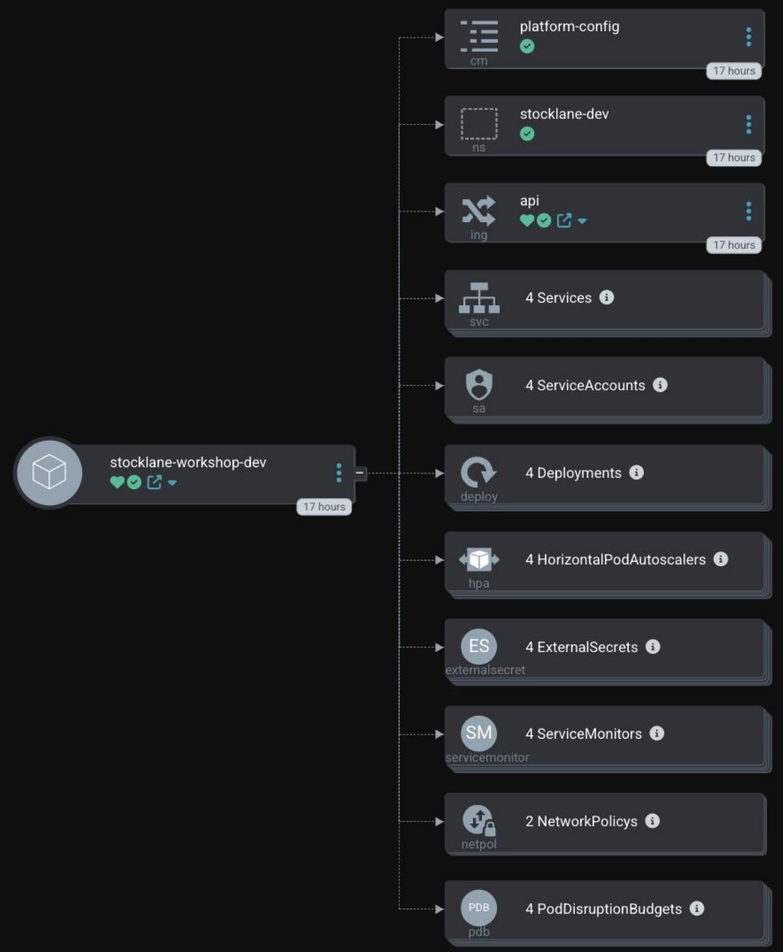
Author: stocklane-release <stocklane-release@users.noreply.github.com> -
Comment: release(dev): c7d2c1b00173 [skip ci]

LAST SYNC
Sync OK to 647bae8

Succeeded 40 minutes ago (Fri Oct 30 2026 17:43:39 GMT+0530)

Author: stocklane-release <stocklane-release@users.noreply.github.com> -
Comment: release(dev): c7d2c1b00173 [skip ci]

80%



Argo CD network view

This is the Argo CD network view. It shows service-to-service relationships and routing; it is deliberately separated from release history.

argo v3.5.6

- Applications
- ApplicationSets
- Settings
- User Info
- Documentation

Resource filters

NAME

NAME

KINDS

KINDS

SYNC STATUS

- Synced 33
- OutOfSync 0

HEALTH STATUS

- Progressing 0
- Suspended 0
- Healthy 21
- Degraded 0
- Missing 0
- Unknown 0

Applications / stocklane-workshop-dev

APPLICATION DETAILS NETWORK

DETAILS | DIFF | SYNC | SYNC STATUS | HISTORY AND ROLLBACK | DELETE | REFRESH

Log out

APP HEALTH
Healthy

SYNC STATUS
Synced to main (647bae8)

Auto sync is enabled.

Author: stocklane-release <stocklane-release@users.noreply.github.com> -
Comment: release(dev): c7d2c1b00173 [skip ci]

LAST SYNC
Sync OK to 647bae8

Succeeded 40 minutes ago (Fri Oct 30 2026 17:43:39 GMT+0530)

Author: stocklane-release <stocklane-release@users.noreply.github.com> -
Comment: release(dev): c7d2c1b00173 [skip ci]

100%

k8s-stocklan-api-01606f23f4-...

17 hours

api

ing

17 hours

forecast

SVC

17 hours

inventory

SVC

17 hours

vision

SVC

17 hours

workshop

SVC

17 hours

Argo CD deployment history

This is the Argo CD history view. Revisions, authors, durations and sync results provide release traceability.

argocd-stocklane.bawantha-rathnayake.tech/applications/argocd/stocklane-workshop-dev

argo v3.5.6

- Applications
- ApplicationSets
- Settings
- User Info
- Documentation

Resource filters

NAME

KINDS

SYNC STATUS
- Synced 33
- OutOfSync 0

HEALTH STATUS
- Progressing 0
- Suspended 0
- Healthy 21
- Degraded 0
- Missing 0
- Unknown 0

Applications / st

DETAILS

APP HEALTH
Healthy

Deployed At:
2 hours ago (Fri Oct 30 2026 16:23:33 GMT+0530)

Time to deploy:
15s

Initiated by:
automated sync policy

Active for:
2h3m

Revision: 647bae8
Authored by: stocklane-release <stocklane-release@users.noreply.github.com>
2 hours ago (Fri Oct 30 2026 16:21:54 GMT+0530)
release(dev): c7d2c1b00173 [skip ci]
GPG signature: -

Source Parameters
URL: https://github.com/bawantha395/workshop-stock-management-system-mlops.git

Deployed At:
2 hours ago (Fri Oct 30 2026 16:01:05 GMT+0530)

Time to deploy:
16s

Initiated by:
automated sync policy

Active for:
22m28s

Revision: f5fd561
Authored by: stocklane-release <stocklane-release@users.noreply.github.com>
2 hours ago (Fri Oct 30 2026 16:00:39 GMT+0530)
release(dev): 10fddf11b800 [skip ci]
GPG signature: -

Source Parameters
URL: https://github.com/bawantha395/workshop-stock-management-system-mlops.git

Deployed At:
3 hours ago (Fri Oct 30 2026 15:18:08 GMT+0530)

Time to deploy:
16s

Initiated by:
automated sync policy

Active for:
42m57s

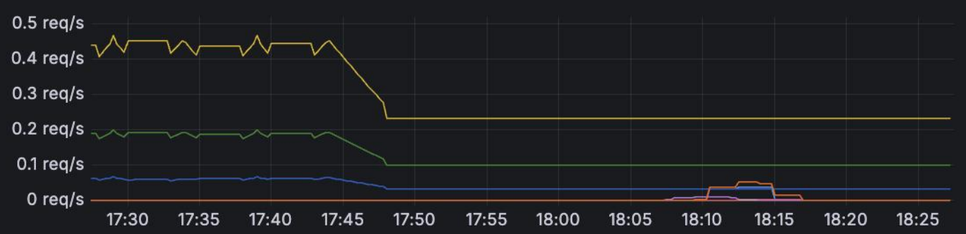
Revision: e9c9936
Authored by: stocklane-release <stocklane-release@users.noreply.github.com>
3 hours ago (Fri Oct 30 2026 15:15:14 GMT+0530)
release(dev): 26dcaf2e4157 [skip ci]
GPG signature: -

Source Parameters
URL: https://github.com/bawantha395/workshop-stock-management-system-mlops.git

Prometheus-backed API dashboard

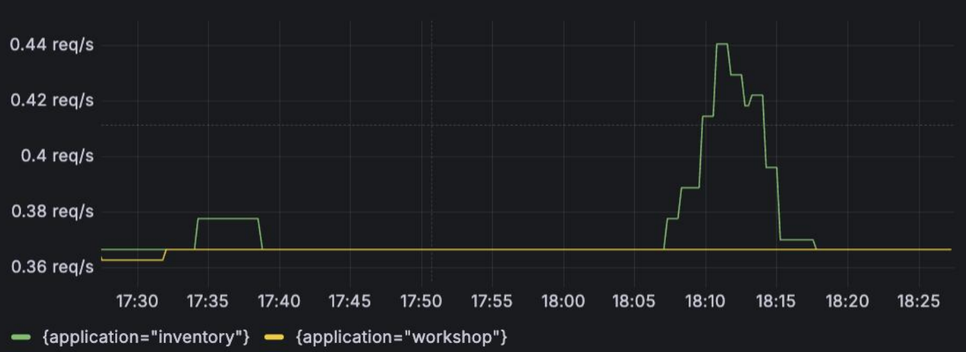
Prometheus metrics visualized in Grafana show API request behaviour, latency and error signals.

Vision HTTP requests / second



Vision p95 request time

Java API requests / second



Vision server errors / second

No data

Embedding cache hit ratio

Application pod restarts

No data

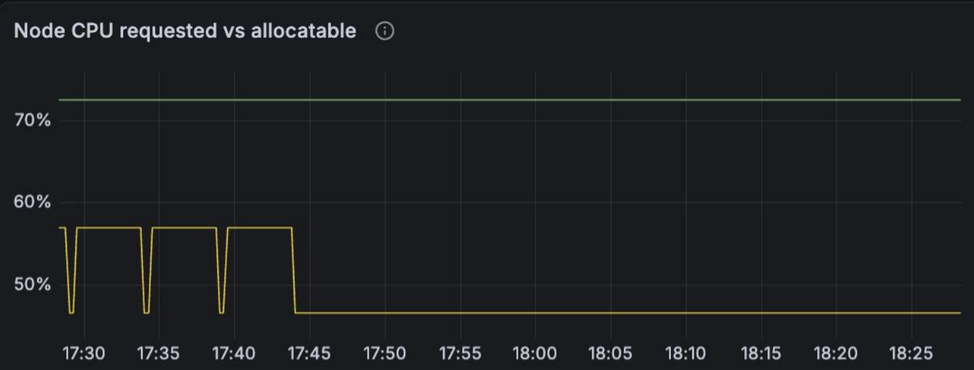
Grafana EKS dashboard

The EKS dashboard provides operational visibility into node CPU, memory, capacity, pods and filesystem usage.

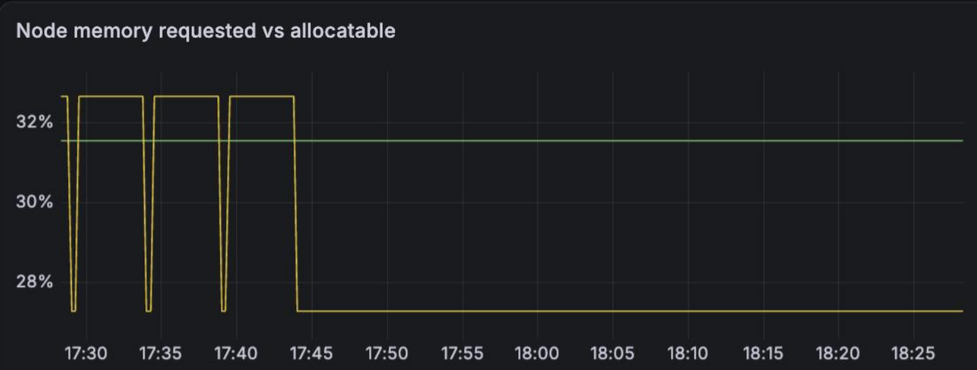
Node CPU usage

Node memory in use (working set)

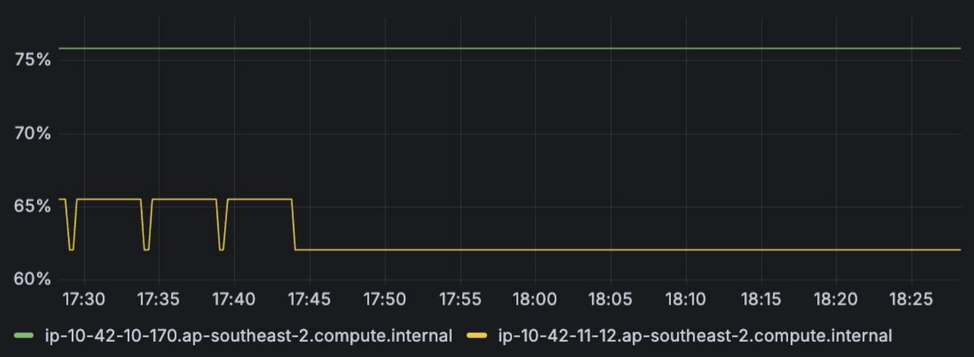
Node CPU requested vs allocatable



Node memory requested vs allocatable



Pods per node vs node pod limit



Node filesystem usage

04 MLOps

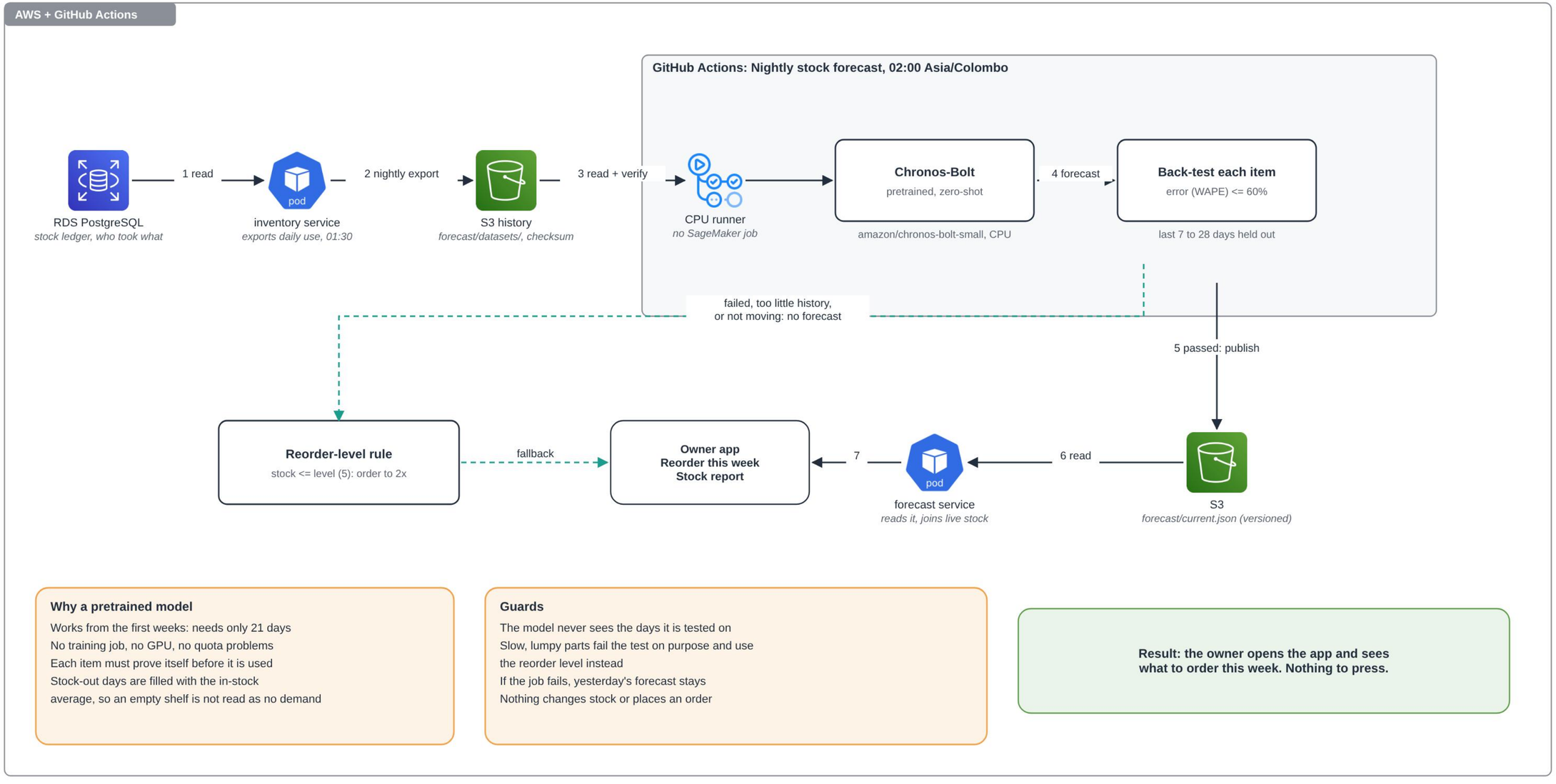
MLOps treats forecasting and photo matching as production model lifecycles rather than isolated notebooks.

- Forecast model: movement history becomes reorder guidance.
- Photo model: an image becomes catalogue candidates.
- SageMaker processing jobs provide visible execution evidence.
- Human confirmation remains part of the production safety boundary.

Forecast model pipeline

Training, evaluation, promotion and scheduled forecast output form the demand-forecast path.

MLOps 1: nightly stock forecast with a pretrained model (no training, no GPU)



Forecast inference lifecycle

Historical movement data is transformed into owner-facing reorder recommendations.

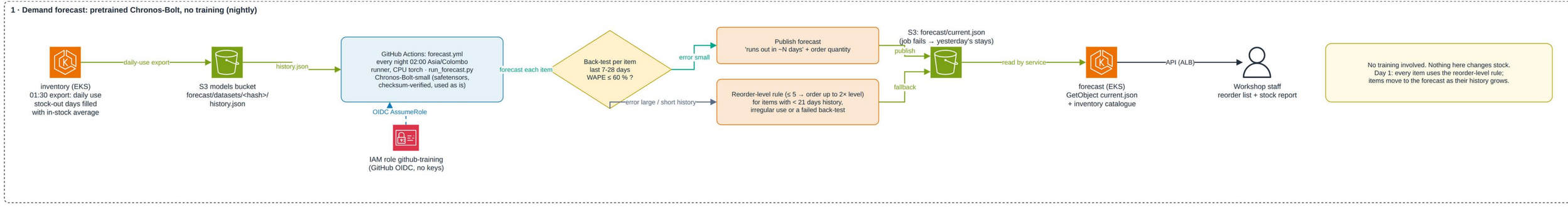

1 · Demand forecast: pretrained Chronos-Bolt, no training (nightly)
inventory (EKS)
01:30 export: daily use
stock-out days filled
with in-stock average
daily-use export
S3 models bucket
forecast/datasets/<hash>/
history.json
history.json
GitHub Actions: forecast.yml
every night 02:00 Asia/Colombo
runner, CPU torch · run_forecast.py
Chronos-Bolt-small (safetensors,
checksum-verified, used as is)
OIDC AssumeRole
IAM role github-training
(GitHub OIDC, no keys)
forecast each item
Back-test per item
last 7-28 days
WAPE ≤ 60 % ?
error small
error large / short history
Publish forecast
'runs out in ~N days' + order quantity
Reorder-level rule (≤ 5 → order up to 2× level)
for items with < 21 days history,
irregular use or a failed back-test
S3: forecast/current.json
(job fails → yesterday's stays)
publish
fallback
read by service
forecast (EKS)
GetObject current.json
+ inventory catalogue
API (ALB)
Workshop staff
reorder list + stock report
No training involved. Nothing here changes stock.
Day 1: every item uses the reorder-level rule;
items move to the forecast as their history grows.

Photo identification model pipeline

Image data, labels, evaluation and model promotion support part identification.

MLOps 2: photo matcher that retrains itself, and label reading with an LLM

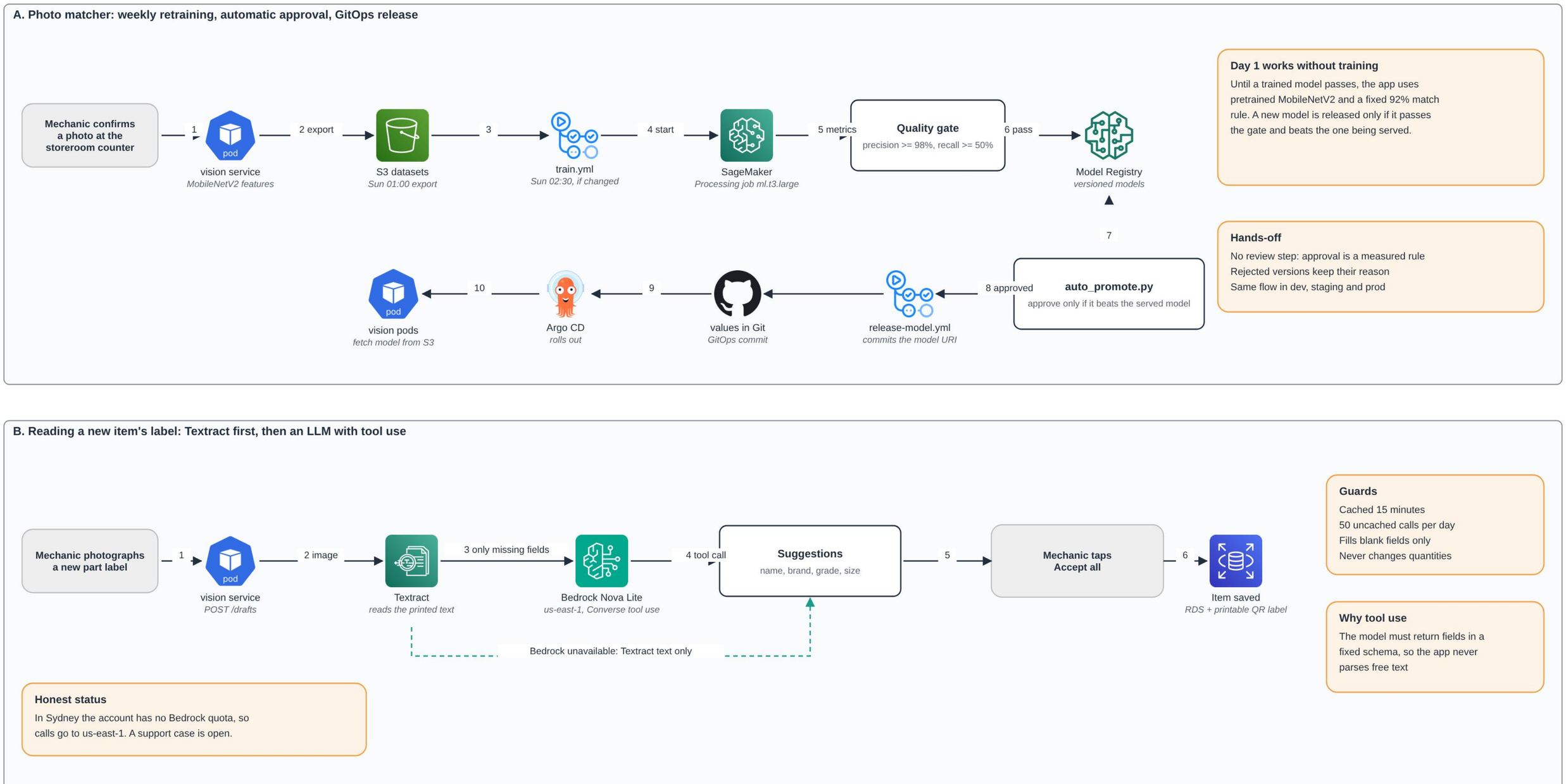
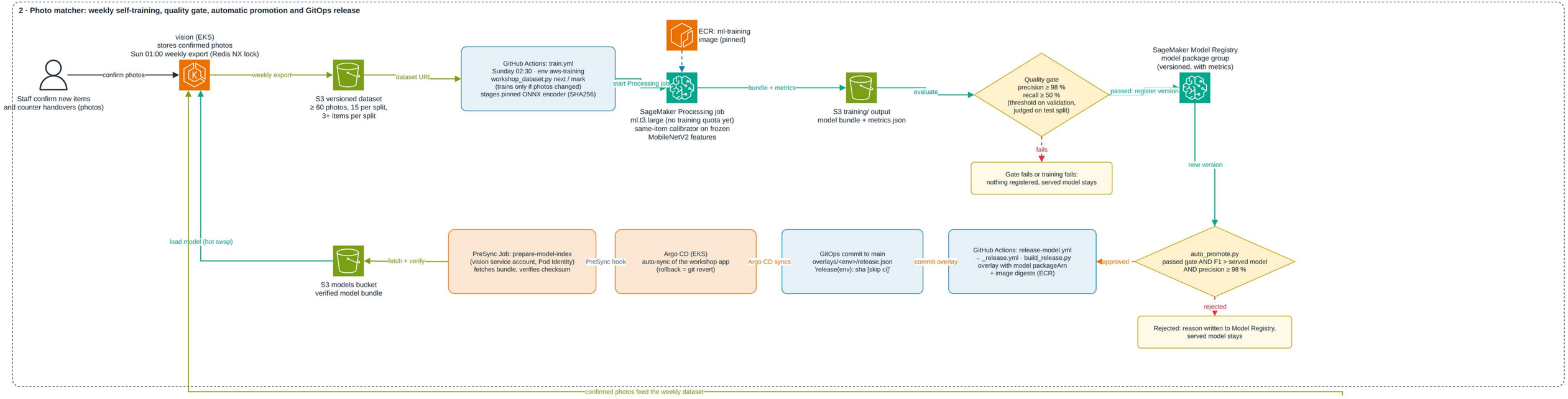


Photo matcher inference lifecycle

A photographed part becomes ranked catalogue candidates that an operator can confirm.


2 · Photo matcher: weekly self-training, quality gate, automatic promotion and GitOps release
Staff confirm new items and counter handovers (photos)
confirm photos
vision (EKS) stores confirmed photos Sun 01:00 weekly export (Redis NX lock)
weekly export
S3 versioned dataset ≥ 60 photos, 15 per split, 3+ items per split
dataset URI
GitHub Actions: train.yml Sunday 02:30 · env aws-training workshop_dataset.py next / mark (trains only if photos changed) stages pinned ONNX encoder (SHA256)
start Processing job
ECR: ml-training image (pinned)
SageMaker Processing job ml.t3.large (no training quota yet) same-item calibrator on frozen MobileNetV2 features
bundle + metrics
S3 training/ output model bundle + metrics.json
evaluate
Quality gate precision ≥ 98 % recall ≥ 50 % (threshold on validation, judged on test split)
passed: register version
SageMaker Model Registry model package group (versioned, with metrics)
fails
Gate fails or training fails: nothing registered, served model stays
new version
auto_promote.py passed gate AND F1 > served model AND precision ≥ 98 %
approved
rejected
Rejected: reason written to Model Registry, served model stays
GitHub Actions: release-model.yml → _release.yml · build_release.py overlay with model packageArn + image digests (ECR)
commit overlay
GitOps commit to main overlays/<env>/release.json 'release(env): sha [skip ci]'
Argo CD syncs
Argo CD (EKS) auto-sync of the workshop app (rollback = git revert)
PreSync hook
PreSync Job: prepare-model-index (vision service account, Pod Identity) fetches bundle, verifies checksum
fetch + verify
S3 models bucket verified model bundle
load model (hot swap)
confirmed photos feed the weekly dataset

SageMaker processing evidence

The SageMaker console shows completed forecast and photo processing jobs. This connects the MLOps diagrams to executed AWS workloads.

customization
Training & tuning jobs
JumpStart model hub New
HyperPod clusters
▶ Training plans
▼ Deployments & inference
Deployable models
Endpoints
Endpoint configurations
Batch transform jobs
Shadow tests
Compilation jobs
Inference Recommender
HyperPod inference
▼ Model governance
Model dashboard
Model cards
▶ AWS Marketplace resources
▼ Data preparation
Processing jobs
▶ Ground Truth
▶ Augmented AI

Tutorials
Documentation

Processing jobs

Actions ▾ | Create processing job

Find processing jobs

1

Name	ARN	Creation time	Duration	Status
stocklane-dev-public-forecast-20261029-122238-d5b0	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-forecast-20261029-122238-d5b0	10/29/2026, 5:52:38 PM	2 minutes	Completed
stocklane-dev-public-photo-20261029-122002-5bd1	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-photo-20261029-122002-5bd1	10/29/2026, 5:50:03 PM	2 minutes	Completed
stocklane-dev-public-forecast-20261029-101055-50fe	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-forecast-20261029-101055-50fe	10/29/2026, 3:40:55 PM	2 minutes	Completed
stocklane-dev-public-photo-20261029-100820-d25e	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-photo-20261029-100820-d25e	10/29/2026, 3:38:21 PM	2 minutes	Completed
stocklane-dev-public-forecast-20261029-092802-5211	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-forecast-20261029-092802-5211	10/29/2026, 2:58:03 PM	2 minutes	Completed
stocklane-dev-public-photo-20261029-092525-b3d7	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-photo-20261029-092525-b3d7	10/29/2026, 2:55:26 PM	2 minutes	Completed
stocklane-processing-test-1793189671	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-processing-test-1793189671	10/28/2026, 5:44:32 PM	a minute	Completed

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05 Application screens

After the model lifecycle is explained, these screens show how an operator enters the application and uses the mobile-friendly workflow.

- The user reaches the application sign-in request.
- Cognito handles authentication.
- The authenticated React home exposes the simple stock actions.
- AI suggestions are reviewed and approved instead of being silently saved.

Application sign-in request

The public application first asks an unauthenticated visitor to sign in before protected inventory actions are available.

stocklane.

WORKSHOP STOCK

- Workspace
- Storeroom counter
- Current stock
- Activity

S

Stock team
Montero Workshop

Workshop inventory

Something needs your attention
Sign in to continue.
Sign in

YOUR STOCK. UNDER CONTROL.

What are we doing today?

Point the camera at a part. That is all.

Receive
Scan or photo what came in

Give a part
Scan the part and confirm the quantity

Find an item

Look up a part, photo or shelf.

Search stock →

Count stock

Compare a physical count with your records.

Check a quantity →

0 catalogue items **0** low-stock items

Cognito hosted sign-in

Amazon Cognito provides the hosted authentication step before the user enters Stocklane.

Sign in with your email and password

Email

name@host.com

Password

Password

Forgot your password?

Sign in

Authenticated Stocklane home

The React home presents Receive, Give a part, Find an item and Count stock in a focused operator workflow.

stocklane.bawantha-rathnayake.tech

stocklane.

WORKSHOP STOCK

Workspace

Storeroom counter

Manage catalogue

Activity

Stock report

YOUR STOCK. UNDER CONTROL.

What are we doing today?

Point the camera at a part. That is all.

REORDER THIS WEEK

2 items to order

Download purchase list

2 to reorder | 2 run out within 3 days | 0 not moving

Acidic Haemodialysis Order soon · 1 in stock	**Order 9** Reorder level
Nirmani Dress Point Order soon · 4 in stock	**Order 6** Reorder level

Forecast updated automatically every night · use through 2026-10-30. Suggestions never change stock.

Receive
Scan or photo what came in

Give a part
Scan the part and confirm the quantity

Find an item

Look up a part, photo or shelf.

Search stock →

Count stock

Compare a physical count with your records.

Check a quantity →

F

Owner workspace
Montero Workshop

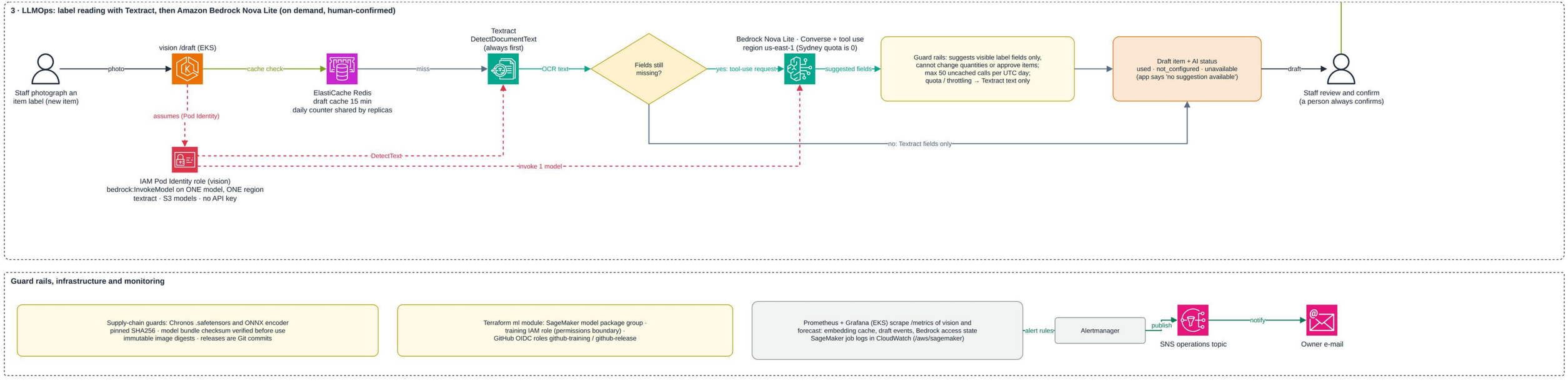
06 LLMOps

LLMOps governs document and label understanding as a reviewable product capability. The system extracts and suggests; it does not silently overwrite catalogue data.

- Textract extracts text from photographed labels.
- Bedrock helps interpret extracted content into catalogue suggestions.
- The actual manufacturer part number remains explicit and required.
- The operator manually approves, edits or rejects the suggestion.

Label-reading architecture

The label-reading flow combines image capture, Textract extraction and Bedrock-assisted interpretation before presenting a suggestion to the operator.


3 · LLMOps: label reading with Textract, then Amazon Bedrock Nova Lite (on demand, human-confirmed)
Staff photograph an item label (new item)
photo
vision /draft (EKS)
cache check
ElastiCache Redis
draft cache 15 min
daily counter shared by replicas
miss
Textract DetectDocumentText (always first)
OCR text
Fields still missing?
yes: tool-use request
Bedrock Nova Lite · Converse + tool use
region us-east-1 (Sydney quota is 0)
suggested fields
Guard rails: suggests visible label fields only, cannot change quantities or approve items; max 50 uncached calls per UTC day; quota / throttling → Textract text only
Draft item + AI status
used · not_configured · unavailable
(app says 'no suggestion available')
draft
Staff review and confirm
(a person always confirms)
no: Textract fields only
assumes (Pod Identity)
DetectText
invoke 1 model
IAM Pod Identity role (vision)
bedrock:InvokeModel on ONE model, ONE region
textract · S3 models · no API key
Guard rails, infrastructure and monitoring
Supply-chain guards: Chronos .safetensors and ONNX encoder pinned SHA256 · model bundle checksum verified before use immutable image digests · releases are Git commits
Terraform ml module: SageMaker model package group · training IAM role (permissions boundary) · GitHub OIDC roles github-training / github-release
Prometheus + Grafana (EKS) scrape /metrics of vision and forecast: embedding cache, draft events, Bedrock access state SageMaker job logs in CloudWatch (/aws/sagemaker)
alert rules
Alertmanager
publish
SNS operations topic
notify
Owner e-mail

Human approval in the Stocklane UI

The UI is the LLMOps control point: suggestions are visible, editable and manually approved. The approved value is then used by the normal catalogue or stock workflow.

- Extract → suggest → review → approve → audit.

stocklane.bawantha-rathnayake.tech

Register a new item

Label readings and AI suggestions are shown for review. Choose Use for each value; nothing is saved automatically.

AI suggestions are not available right now (the AWS Bedrock quota for this account is not approved yet). Showing what the label says and similar items only.

3 suggestions ready for the empty fields — Accept all

Save the item first, then confirm how many came in.

Item name

Suggested: **Mitsubishi Gl-5 Super Hypoid Gear Oil** · Label suggestion Use

Manufacturer / brand

Suggested: **Mitsubishi** · Label suggestion Use

Actual manufacturer part number

Category

Suggested: **Engine oil** · Label suggestion Use

Shelf / location

Counting unit

Piece / sealed unit / defined set

▶ More details

Cancel — Save without suggestions

07 • SECTION

07 AWS infrastructure

The AWS evidence follows the infrastructure diagram from the outer edge to the running application: global edge and identity, VPC, container runtime, data services, messaging, security, AI/ML and observability.

- Every screenshot below is an original user capture from the current folder.
- Generated aws-screens images and earlier PDFs are excluded.
- The service name is stated before each capture so the console evidence is not confused with an architecture diagram.

A • Global edge, DNS and identity

Route 53 resolves public names, the load balancer exposes the application edge, and Cognito supplies user identity.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

Route 53 hosted zone

DNS records connect public application, API and platform endpoints.

Route 53

Dashboard
Hosted zones
Health checks
Profiles

▼ Global Resolver
Global resolvers New
Shared DNS views New

▼ VPC Resolver
VPCs
Inbound endpoints
Outbound endpoints
Rules
Query logging
Outposts

▼ Domains
Registered domains
Requests

▼ IP-based routing
CIDR collections

▼ Traffic flow
Traffic policies
Policy records

DNS Firewall
Application Recovery Controller

Public bawantha-rathnayake.tech Info

Delete zone | Test record | Configure query logging

Hosted zone details

Edit hosted zone

Records (10) | Accelerated recovery | DNSSEC signing | Hosted zone tags (0)

Records (10) Info

Delete record | Import zone file | Create record

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value | Type | Routing p... | Alias | 1

Record name	Type	Routin...	Differ...	Alias
bawantha-rathnayake.tech	NS	Simple	-	No
bawantha-rathnayake.tech	SOA	Simple	-	No
argocd-stocklane.bawantha-rathnayake.tech	A	Simple	-	Yes
_dd976b6d9e8daebe8816434bb4998797.argocd-stocklane.bawantha-rathnayake.t…	CNAME	Simple	-	No
grafana-stocklane.bawantha-rathnayake.tech	A	Simple	-	Yes
_2715d100bfc104d49d710f3b5c1534fc.grafana-stocklane.bawantha-rathnayake.t…	CNAME	Simple	-	No
stocklane.bawantha-rathnayake.tech	A	Simple	-	Yes
_45771f7f67bf431675e48f2f8e320001.stocklane.bawantha-rathnayake.tech	CNAME	Simple	-	No
api.stocklane.bawantha-rathnayake.tech	A	Simple	-	Yes
_5d3af9978a29f8301dad61fcffeede29.api.stocklane.bawantha-rathnayake.tech	CNAME	Simple	-	No

0 records selected

Select a record to see its details

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Amazon Cognito user pool

The user pool provides the authentication boundary for protected Stocklane workflows.

Amazon Cognito

User pools
Identity pools

User pools | Provisioned limits New

New from Amazon Verified Permissions! Cognito user group authorization for API Gateway
You can now create group-aware authorization policies for your APIs with Amazon Verified Permissions, a fine-grained authorization service for applications. Learn more

Go to Amazon Verified Permissions

User pools (1) Info

Delete | Create user pool

View and configure your user pools. User pools are directories of federated and local user profiles. They provide authentication options for your users.

Search user pools by name or ID

1

User pool name	User pool ID	Regions	Created time	Last updated time
stocklane-dev	ap-southeast-2_kESdbq2P6	Single-Region	17 hours ago	17 hours ago

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Application Load Balancers

The EC2 load-balancer view shows active internet-facing ALBs used by the Kubernetes platform.

Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups
Trust Stores

▼ Auto Scaling
Auto Scaling Groups

Settings

Load balancers (2) What's new?

Actions ▼ | Create load balancer ▼

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

1

Name	State	Type	Scheme	IP address type	VPC ID
k8s-stocklanedevadmin-640ae9aa82	Active	application	Internet-facing	IPv4	vpc-0221f7212e3647ffb
k8s-stocklan-api-01606f23f4	Active	application	Internet-facing	IPv4	vpc-0221f7212e3647ffb

0 load balancers selected

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Target groups

Kubernetes-managed IP target groups connect ALB listeners to EKS services.

Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New

Images
AMIs
AMI Catalog

Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Load Balancing
Load Balancers
Target Groups
Trust Stores

Auto Scaling
Auto Scaling Groups

Settings

Target groups (6) Info | What's new?

Actions ▼ Create target group

Filter target groups

1

Name	ARN	Port	Protocol	Target type	Load balancer
k8s-argocd-argocdse-1b7a98e730	arn:aws:elasticloadbalancin…	8080	HTTPS	IP	k8s-stocklanedev
k8s-monitori-grafana-bbadb45c9c	arn:aws:elasticloadbalancin…	1	HTTP	IP	k8s-stocklanedev
k8s-stocklan-forecast-829f7a8543	arn:aws:elasticloadbalancin…	1	HTTP	IP	k8s-stocklan-api-
k8s-stocklan-inventor-f47fbd2a91	arn:aws:elasticloadbalancin…	1	HTTP	IP	k8s-stocklan-api-
k8s-stocklan-vision-0a2f8a69b0	arn:aws:elasticloadbalancin…	1	HTTP	IP	k8s-stocklan-api-
k8s-stocklan-workshop-d05236995c	arn:aws:elasticloadbalancin…	1	HTTP	IP	k8s-stocklan-api-

0 target groups selected

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B • VPC and network foundation

The VPC provides public and private subnets, routing, internet egress and private AWS service connectivity.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

Stocklane VPC

The stocklane-dev VPC and CIDR establish the network boundary.

VPC dashboard

Filter by VPC:
Choose a VPC

AWS Global View

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Security
Network ACLs
Security groups

PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services

Your VPCs

VPCs | VPC encryption controls

Your VPCs (1/2) Info

Last updated 1 minute ago | Actions | Create VPC

Find VPCs by attribute or tag

1

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR
stocklane-dev	vpc-0221f7212e3647ffb	Available	–	–	Off	10.42.0.0/16
–	vpc-0079eab69c25af555	Available	–	–	Off	172.31.0.0/16

vpc-0221f7212e3647ffb / stocklane-dev

Details | Resource map | CIDRs | Flow logs | Tags | Related resources | Integrations

IPv4 CIDRs Info

Edit CIDRs

Address family	CIDR	Status
IPv4	10.42.0.0/16	Associated

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Public and private subnets

The subnet view shows separate public and private subnets for controlled placement.

VPC dashboard

Filter by VPC:
Choose a VPC

AWS Global View

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Security
Network ACLs
Security groups

PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services

Subnets (7) Info

Last updated 1 minute ago | Actions | Create subnet

Find subnets by attribute or tag

1

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
–	subnet-0087fffda1b524cae	Available	vpc-0079eab69c25af555	Off	172.31.16.0/20
–	subnet-0e83cf3eda7dc3aab	Available	vpc-0079eab69c25af555	Off	172.31.0.0/20
–	subnet-040826a229c4cf3da	Available	vpc-0079eab69c25af555	Off	172.31.32.0/20
stocklane-dev-private-0	subnet-0082379910b54c1d0	Available	vpc-0221f7212e3647ffb | stock...	Off	10.42.10.0/24
stocklane-dev-public-1	subnet-0129744129d0ba7da	Available	vpc-0221f7212e3647ffb | stock...	Off	10.42.1.0/24
stocklane-dev-public-0	subnet-0fa1dc46fe8527f86	Available	vpc-0221f7212e3647ffb | stock...	Off	10.42.0.0/24
stocklane-dev-private-1	subnet-0f3af3bd90fa23362	Available	vpc-0221f7212e3647ffb | stock...	Off	10.42.11.0/24

Select a subnet

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Internet Gateway

The Internet Gateway supports public-subnet connectivity.

VPC dashboard

Filter by VPC:
Choose a VPC

AWS Global View

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Security
Network ACLs
Security groups

PrivateLink and Lattice
Getting started
Endpoints
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Lattice services

igw-0c8b41b7dbd35e0c7

Actions

Details Info

Internet gateway ID	State	VPC ID	Owner
igw-0c8b41b7dbd35e0c7	Attached	vpc-0221f7212e3647ffb | stocklane-dev	721274598579

Tags (3)

Manage tags

Search tags

1

Key	Value
Project	stocklane-dev
ManagedBy	Terraform
Environment	dev

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NAT Gateway

The NAT Gateway gives private workloads controlled outbound access.

VPC dashboard

Filter by VPC:
Choose a VPC

AWS Global View

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Security
Network ACLs
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PrivateLink and Lattice
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Endpoints
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Lattice services

nat-07b609790b5985c67

Actions

Details

NAT gateway ID
nat-07b609790b5985c67

NAT gateway ARN
arn:aws:ec2:ap-southeast-2:721274598579:natgateway/nat-07b609790b5985c67

VPC
vpc-0221f7212e3647ffb / stocklane-dev

Connectivity type
Public

Primary public IPv4 address
54.252.162.246

Subnet
subnet-0fa1dc46fe8527f86 / stocklane-dev-public-0

State
Available

Primary private IPv4 address
10.42.0.12

Created
Friday, 30 October 2026 at 00:42:22 GMT+5:30

State message Info
–

Primary network interface ID
eni-0ed8e5781a8a4405c

Deleted
–

Secondary IPv4 addresses | Monitoring | Tags

Secondary IPv4 addresses

Edit secondary IPv4 address associations

Search

1

Private IPv4 address	Allocation ID	Association ID	Public IPv4 address	Network interf

Secondary IPv4 addresses are not available for this nat gateway.

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S3 VPC endpoint

The gateway endpoint provides VPC access to S3 without sending that traffic through a public internet path.

VPC dashboard

Filter by VPC:
Choose a VPC

AWS Global View

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Security
Network ACLs
Security groups

PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services

vpce-0ff83ecb7ab23c8a5

Actions

Details

Endpoint ID
vpce-0ff83ecb7ab23c8a5

VPC ID
vpc-0221f7212e3647ffb (stocklane-dev)

DNS record IP type
service-defined

Private DNS specified domains
–

Status
Available

Status message
–

IP address type
ipv4

Creation time
Friday, 30 October 2026 at 00:44:04 GMT+5:30

Service name
com.amazonaws.ap-southeast-2.s3

Service region
ap-southeast-2

Endpoint type
Gateway

Private DNS names enabled
No

Private DNS preference
–

Route tables | Policy | Tags

Route tables (2)

Manage route tables

Search

1

Name	Route Table ID	Main	Associated Id
–	rtb-0d0bc71b803484d88	No	subnet-0f3af3bd90fa23362 (stocklane-dev-private-1)
–	rtb-0979df2c50ab58cb2	No	subnet-0082379910b54c1d0 (stocklane-dev-private-0)

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C • Compute and container registry

EKS runs the Spring Boot and FastAPI microservices; ECR stores the container images used by deployments. Kubernetes is covered here as part of the AWS runtime, not as a separate section.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

EKS cluster

The EKS console shows the running cluster, nodes, pods, add-ons and identity configuration.

Amazon Elastic Kubernetes Service

Dashboard Updated
Clusters Updated

▼ Cluster: stocklane-dev
- Cluster monitoring
- Resources
- Compute
- Add-ons
- Capabilities
- Access
- Backups

▼ Settings
- Dashboard settings
- Console settings

▼ Amazon EKS Anywhere
- Enterprise Subscriptions

▼ Related services
- Amazon ECR
- AWS Batch

- Documentation
- EKS workshops

Cluster resources

Last updated
October 30, 2026, 17:42 (UTC+05:30)

Create

Pods	Nodes	Node groups	Fargate profiles	Add-ons 3	Capabilities
41	2	1	0	5	0

IAM access entries	Pod Identities	OIDC identity providers
4	10	0

Configuration Updated | Update history | Certificate authority

Cluster configuration

Manage

EKS Auto Mode | Info
Disabled

Upgrade policy
Extended support

Authentication mode
EKS API

Scaling tier | Info
Standard View usage

Deletion protection | Info
Disabled

ARC Zonal shift | Info
Disabled

Encryption key | Info
Customer managed

KMS key ID
ef2a7884-7f9d-44e5-8f68-93f0858f5de9

Networking

Manage

VPC | Info
vpc-0221f7212e3647ffb

Cluster IP address family | Info
IPv4

Service IPv4 range | Info
172.20.0.0/16

Subnets
subnet-0082379910b54c1d0
subnet-0f3af3bd90fa23362

Cluster security group | Info
sg-0501ea9dc4d2d28d4

Additional security groups
None

API server endpoint access | Info
Public and private

Egress mode | Info
AWS managed

Public access source allowlist
112.135.67.224/32

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EKS workload resources

The EKS resources view shows running pods, DaemonSets, Deployments, ReplicaSets, StatefulSets and Jobs.

Amazon Elastic Kubernetes Service

Dashboard Updated
Clusters Updated

▼ Cluster: stocklane-dev
- Cluster monitoring
- Resources
- Compute
- Add-ons
- Capabilities
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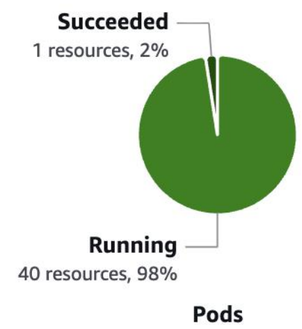
Resources

Workload Status

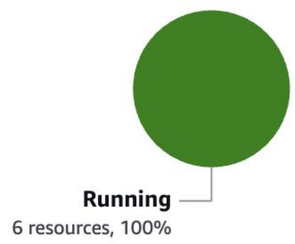
Overview of workload resources grouped by status.

Last updated
October 30, 2026, 17:44 (UTC+05:30)

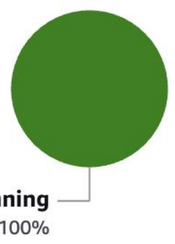
Filter namespaces



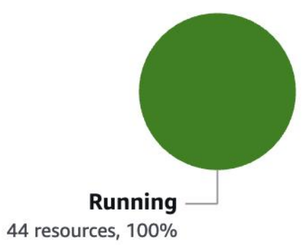
Pods



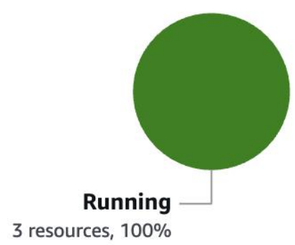
DaemonSets



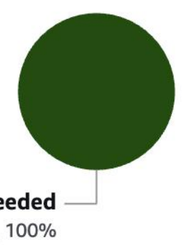
Deployments



ReplicaSets



StatefulSets



Jobs

Resource types

Filter resource type

Workloads: Pods (41)

Pod is the smallest and simplest Kubernetes object. A Pod represents a set of running containers on your cluster. Learn more

Last updated
October 30, 2026, 17:42 (UTC+05:30)

View details

Filter namespaces

Filter Pods by name

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EKS runtime architecture

The cluster diagram explains how ingress, Services, Deployments, autoscalers, policies and monitoring run the application workloads.

Stocklane - full AWS architecture (all components and connections, one environment)

user traffic · service-to-service · async events · data / storage · auth / secrets (dashed) · logs / alerts · GitOps (dashed) · CI/CD (dashed) · ML / AI

AWS Cloud (one region)

VPC (public + private + data subnets, 2-3 AZs)

Public subnets

Route 53 hosted zone
DNS validation

ACM certificates web · api · admin
TLS cert

Internet gateway

NAT gateway + Elastic IP (1 per AZ in prod)

Application Load Balancer (ACM cert, HTTPS 443)

CloudFront + Origin Access Control

S3 web bucket React build

Cognito user pool web client · domain staff group · MFA

Staff & owner phone / browser

Operator kubectl / console (IP allow-list)

Private subnets (worker nodes, 2-3 AZs)

Amazon EKS cluster - managed control plane, KMS-encrypted secrets, endpoint restricted to operator CIDR

ns: stocklane-env

Ingress (ALB, IP targets) /api/inventory · /api/workshop /api/vision · /api/forecast

ConfigMap + ExternalSecret

NetworkPolicies default-deny + allow

Job: database-bootstrap (schemas)

Job: prepare-model-index (Argo PreSync)

Deployments (ClusterIP)

workshop :8082 Spring Boot

inventory :8081 Spring Boot

vision :8083 FastAPI · MobileNetV2

forecast :8084 FastAPI

ns: external-secrets · cert-manager

cert-manager

External Secrets Operator (SecretStore → Secrets Manager)

ns: argocd

Applications: platform · monitoring · workshop (Kustomize overlays, auto-sync)

Argo CD server · repo · controller

ns: kube-system (add-ons)

metrics-server

CoreDNS

kube-proxy

VPC CNI (NetworkPolicy on)

EBS CSI driver

Pod Identity Agent

AWS Load Balancer Controller

Pod Identity roles (permissions boundary): inventory · workshop · vision · forecast · db-bootstrap external-secrets · LB controller · fluent-bit · alertmanager

ns: monitoring

Prometheus Operator

Grafana (Prometheus datasource)

kube-state-metrics

Prometheus + ServiceMonitors

Alertmanager

PrometheusRule alerts · scrapes services (/actuator/prometheus, /metrics), kubelet, kube-state-metrics

Worker nodes

Managed node group launch template · IMDSv2

EBS gp3 volumes (Prometheus, Grafana PVCs)

ns: amazon-cloudwatch

Fluent Bit DaemonSet (container logs)

CodeBuild (platform-bootstrap) private in-VPC runner

Security group: egress 443 only, EKS API ingress from runner

Private data subnets

RDS PostgreSQL 17 subnet group · Multi-AZ in prod KMS-encrypted

ElastiCache Redis 7.1 replication group · TLS failover + Multi-AZ

Data security group: 5432 / 6379 only from node SG

Route tables: public → IGW · private → NAT S3 gateway endpoint on private tables

S3 gateway VPC endpoint

Data, messaging and AI services used by the pods

SQS events queue outbox events

SQS DLQ

S3 private bucket photos · models · datasets forecast/current.json

Bedrock (Nova Lite) label reading

Textract label OCR

Secrets and encryption

Secrets Manager database · shared · grafana

KMS CMK RDS · S3 · secrets · EKS secrets

Observability and alerting

CloudWatch Logs EKS control plane

KMS key alerts topic

CloudTrail recent changes

AWS Budgets monthly cost

SNS operations topic (TLS only, KMS)

CloudWatch Logs application log group

CloudWatch alarms errors · queue age · DLQ

EventBridge rule alarm state change

Lambda alarm RCA evidence + summary

DynamoDB alarm_rca (dedupe)

CI/CD identity, registry and ML

S3 automation bucket deployment metadata

S3 Terraform state (CI foundation)

CloudFormation CI foundation stack

ECR image repositories

IAM roles (3) platform · inference · training

IAM OIDC provider GitHub Actions

SageMaker training + processing jobs

SageMaker model package group (registry)

Developer git push

GitHub repo (main)

Actions runners: deploy.yml forecast.yml (nightly) train.yml (weekly) release-model.yml destroy.yml

OIDC identity

Owner e-mail (SNS subscriber)

HTTPS API · DNS · HTTPS app · sign in + MFA · kubectl (EKS API, access entry) · SQL (inventory, workshop) · cache (inventory, TLS) · TLS cert · ALIAS app · HTTPS 443 · internet egress · egress (ECR, AWS APIs) · ALB target groups · verify JWT (JWKS) · origin (OAC) · path routing · HTTP · HTTP :8081 · reconciles ALB · ReceiveMessage · SendMessage (outbox relay) · redrive · PutObject datasets · photos · model · Converse · DetectText · GetObject current.json · control-plane logs · container logs · image pull · scrape · alert rules · publish alerts · LookupEvents · e-mail · RCA summary · alarm / ok actions · metric filter · state change · invoke · dedupe / daily slot · Logs Insights · for datasets / artifacts3 write) · sync secrets · decrypt · encrypt · queue age + DLQ depth · creates · deploy metadata · push images (digest) · AssumeRole · start training · register package · OIDC token · pulls overlay (GitOps) · install Argo CD · publish frontend · invalidate cache · start build (bootstrap)

AWS Load Balancer Controller

The controller reconciles Kubernetes Ingress and Service resources into AWS load balancers and target groups using Pod Identity.

Amazon Elastic Kubernetes Service

Dashboard Updated
Clusters Updated

▼ Cluster: stocklane-dev
- Cluster monitoring
- Resources
- Compute
- Add-ons
- Capabilities
- Access
- Backups

▼ Settings
- Dashboard settings
- Console settings

▼ Amazon EKS Anywhere
- Enterprise Subscriptions

▼ Related services
- Amazon ECR
- AWS Batch

- Documentation
- EKS workshops

aws-load-balancer-controller-595cfcc9cf-4hc8v

Structured view | Raw view

Details

Status
Running

Created
19 hours ago

Namespace
kube-system

Controlled by
ReplicaSet/aws-load-balancer-controller-595cfcc9cf

Last transition time
19 hours ago

Node
ip-10-42-10-170.ap-southeast-2.compute.internal

Pod IP
10.42.10.105

Containers (1)

▶ **controller**

Labels (6)

1 2

Key	Value
app.kubernetes.io/component	controller
app.kubernetes.io/name	aws-load-balancer-controller
eks.amazonaws.com/pod-identity	enabled
pod-template-hash	595cfcc9cf
topology.kubernetes.io/region	ap-southeast-2

Annotations (0)

1

Key	Value

No annotations
No annotations have been defined for this resource.

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Amazon ECR repositories

ECR stores separate encrypted repositories for inventory, workshop, vision, forecast, operations and ML training images.

Amazon Elastic Container Registry

▼ Private registry
Repositories
Features & Settings

▼ Public registry
Repositories
Settings

ECR public gallery
Amazon ECS
Amazon EKS

Getting started
Documentation

Private repositories (6)

View push commands | Delete | Actions | Create repository

Search by repository substring

Repository name	URI	Created at	Tag immutability	Encryption type
stocklane-dev/forecast	721274598579.dkr.ecr.ap-southeast-2.amazonaws.com/stocklane-dev/forecast	30 October 2026, 00:42:39 (UTC+05.5)	Immutable	KMS
stocklane-dev/inventory	721274598579.dkr.ecr.ap-southeast-2.amazonaws.com/stocklane-dev/inventory	30 October 2026, 00:42:38 (UTC+05.5)	Immutable	KMS
stocklane-dev/ml-training	721274598579.dkr.ecr.ap-southeast-2.amazonaws.com/stocklane-dev/ml-training	30 October 2026, 00:42:37 (UTC+05.5)	Immutable	KMS
stocklane-dev/operations	721274598579.dkr.ecr.ap-southeast-2.amazonaws.com/stocklane-dev/operations	30 October 2026, 00:42:37 (UTC+05.5)	Immutable	KMS
stocklane-dev/vision	721274598579.dkr.ecr.ap-southeast-2.amazonaws.com/stocklane-dev/vision	30 October 2026, 00:42:39 (UTC+05.5)	Immutable	KMS
stocklane-dev/workshop	721274598579.dkr.ecr.ap-southeast-2.amazonaws.com/stocklane-dev/workshop	30 October 2026, 00:42:38 (UTC+05.5)	Immutable	KMS

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EBS persistent volumes

The EBS view shows gp3 volumes used by stateful Kubernetes workloads.

- Instances
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - UltraServers New
  - Capacity Manager New
- Images
  - AMIs
  - AMI Catalog
- Elastic Block Store
  - Volumes
  - Snapshots
  - Lifecycle Manager
- Network & Security
  - Security Groups
  - Elastic IPs
  - Placement Groups
  - Key Pairs
  - Network Interfaces
- Load Balancing
  - Load Balancers
  - Target Groups

Volumes (5) Info

Last updated less than a minute ago | Recycle Bin | Actions | Create volume

Saved filter sets: Choose filter set | Search

1

Name	Volume ID	Type	Size	IOPS	Throughput	Owner ID	Snapshot ID
	vol-0e08b628c098eb46f	gp3	30 GiB	3000	125	721274598579	snap-0579775...
stocklane-dev-dynamic-pvc-994331f7-...	vol-09cf10c937bff5ae6	gp3	20 GiB	3000	125	721274598579	-
stocklane-dev-dynamic-pvc-588326a4-...	vol-04bcbfed8ac2c5f41	gp3	5 GiB	3000	125	721274598579	-
stocklane-dev-dynamic-pvc-f66fdaa3-3...	vol-0fdb13af54fc2e986	gp3	1 GiB	3000	125	721274598579	-
	vol-02795a12f41b8dd13	gp3	30 GiB	3000	125	721274598579	snap-0579775...

Fault tolerance for all volumes in this Region

Snapshot summary

Last updated on Fri, Oct 30, 2026, 08:09:19 PM (GMT+05:30)

Snapshots to back up volumes can be manually created, or created through automation such as Amazon Data Lifecycle Manager, AWS Backup, custom scripts, or third-party tools.

Volumes backed up in the last 24 hours / Total # volumes
0 / 5

Data Lifecycle Manager default policy for EBS Snapshots status
No default policy set up | Create policy

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D • S3 object storage

S3 stores media, model artefacts, Terraform state and automation outputs with separate buckets for platform responsibilities.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

S3 buckets

The bucket list shows automation, media, models, Terraform state and web assets.

Amazon S3

▼ Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets

▼ Files
File systems

▼ Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer

▼ Storage management and insights
Storage Lens
Batch Operations

Account and organization settings
▶ AWS Marketplace for S3

Buckets

General purpose buckets | All AWS Regions | Directory buckets

General purpose buckets (5) Info

Copy ARN | Empty | Delete | Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

1

Name	AWS Region	Creation date
stocklane-dev-automation-721274598579	Asia Pacific (Sydney) ap-southeast-2	October 30, 2026, 00:42:15 (UTC+05:30)
stocklane-dev-media-721274598579	Asia Pacific (Sydney) ap-southeast-2	October 30, 2026, 00:42:16 (UTC+05:30)
stocklane-dev-models-721274598579	Asia Pacific (Sydney) ap-southeast-2	October 30, 2026, 00:42:17 (UTC+05:30)
stocklane-dev-tfstate-721274598579-ap-southeast-2	Asia Pacific (Sydney) ap-southeast-2	October 25, 2026, 23:53:31 (UTC+05:30)
stocklane-dev-web-721274598579	Asia Pacific (Sydney) ap-southeast-2	October 30, 2026, 00:42:16 (UTC+05:30)

▶ Account snapshot Info

Updated daily

View dashboard

Storage Lens provides visibility into storage usage and activity trends.

▶ External access summary Info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

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S3 model artefacts

The models bucket contains forecast and release artefacts used by the MLOps lifecycle.

Amazon S3

Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets

Files
File systems

Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer

Storage management and insights
Storage Lens
Batch Operations

Account and organization settings
AWS Marketplace for S3

stocklane-dev-models-721274598579 Info

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems | Access Points

Objects (2)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

Find objects by prefix | Show versions | 1

Name	Type	Last modified	Size	Storage class
forecast/	Folder	-	-	-
releases/	Folder	-	-	-

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E • Data services

PostgreSQL is the transactional source of truth and Redis provides disposable application caching. Neither replaces the authoritative inventory record.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

PostgreSQL database

The RDS console shows the stocklane PostgreSQL database, encryption and CloudWatch log integration.

Aurora and RDS

Dashboard
Databases
Query editor
CloudWatch Database Insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Capacity reservations New
Proxies

Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations

Events
Event subscriptions

Recommendations 0
Certificate update

stocklane-dev

Modify | Actions

Summary

DB identifier	Status	Role	Engine	Recommendations
stocklane-dev	Available	Instance	PostgreSQL	
CPU 3.47%	**Class** db.t4g.micro	**Current activity** 22 Connections	**Region & AZ** ap-southeast-2b	

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations

Instance

Configuration

DB instance ID
stocklane-dev

Engine version
17.9

RDS Extended Support
Enabled

DB name
stocklane

License model
Postgresql License

Option groups
default:postgres-17 In sync

Amazon Resource Name (ARN)
arn:aws:rds:ap-southeast-2:721274598579:db:stocklane-dev

Instance class

Instance class
db.t4g.micro

vCPU
2

RAM
1 GB

Availability

Master username
stocklane_admin

IAM DB authentication
Not enabled

Multi-AZ
No

Secondary Zone

Primary storage

Encryption
Enabled

AWS KMS key
stocklane-dev

Storage type
General Purpose SSD (gp2)

Storage
20 GiB

Provisioned IOPS
-

Storage throughput
-

Storage autoscaling
Disabled

Monitoring

Monitoring type
Database Insights - Standard

Database counter metrics
Disabled

Enhanced Monitoring
Disabled

Logs published to Cloudwatch
PostgreSQL log
Upgrade log

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Redis cache

ElastiCache Redis provides an available, encrypted, multi-node application cache and is explicitly not the inventory authority.

Amazon ElastiCache

Dashboard

Resources
Valkey caches
Memcached caches
Redis OSS caches
Global datastores
Reserved nodes
Backups

Configurations
Subnet groups
Parameter groups
User management
User group management

Migrations New
Events
Service updates

ElastiCache cluster client
Documentation
Amazon MemoryDB

stocklane-dev Info

Upgrade to Valkey | Modify | Actions

Cluster details

Cluster name stocklane-dev	**Description** Disposable application cache, never inventory authority	**Node type** cache.t4g.micro	**Status** Available
Engine Redis	**Engine version** 7.1.0	**Global datastore** -	**Global datastore role** -
Update status Up to date	**Cluster mode** Disabled	**Shards** 1	**Number of nodes** 2
Data tiering Disabled	**Multi-AZ** Enabled	**Auto-failover** Enabled	**Encryption in transit** Enabled
Encryption at rest Enabled	**Parameter group** default.redis7	**Outpost ARN** -	**Transit encryption mode** Required
Configuration endpoint -	**Primary endpoint** master.stocklane-dev.g3lygp.apse2.cache.amazonaws.com:6379	**Reader endpoint** replica.stocklane-dev.g3lygp.apse2.cache.amazonaws.com:6379	**ARN** arn:aws:elasticache:ap-southeast-2:721274598579:replicationgroup:stocklane-dev
Data migration No active migrations			

Connectivity and security - *new* | Nodes | Metrics | Logs | Maintenance and backups | Service updates | Tags

Nodes (2)

Manage tags | Failover primary | Promote | Reboot node | Delete node | Add node

Find nodes

1

Node name	Status	Current role	Endpoint	ARN
stocklane-dev-001	Available	primary	stocklane-dev-001.stocklane-dev.g3lygp.apse2.cache.amazonaws.com:6379	arn:aws:elasticache:ap-southeast-2:721274598579
stocklane-dev-002	Available	replica	stocklane-dev-002.stocklane-dev.g3lygp.apse2.cache.amazonaws.com:6379	arn:aws:elasticache:ap-southeast-2:721274598579

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F • SQS asynchronous messaging

SQS decouples catalogue and stock work, with dead-letter queues for failed messages.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

SQS queues

The queue list shows catalogue and stock queues plus their DLQs, encrypted with KMS.

Amazon SQS

Queues

Queues (4)

Edit | Delete | Send and receive messages | Actions | Create queue

Search queues by prefix

1

Name	Type	Created	Messages available	Messages in flight	Encryption
stocklane-dev-catalogue	Standard	2026-10-30T00:42+05:30	0	0	AWS Key Management Service key (SSE-KMS)
stocklane-dev-catalogue-dlq	Standard	2026-10-30T00:42+05:30	0	0	AWS Key Management Service key (SSE-KMS)
stocklane-dev-stock	Standard	2026-10-30T00:42+05:30	0	0	AWS Key Management Service key (SSE-KMS)
stocklane-dev-stock-dlq	Standard	2026-10-30T00:42+05:30	0	0	AWS Key Management Service key (SSE-KMS)

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G • Keys and secrets

KMS encryption and Secrets Manager keep data and credentials outside source code and container images.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

Secrets Manager

Secrets are stored as managed secrets for database, platform and observability integrations.

Secrets

Recommended actions | Store a new secret

Filter secrets by name, description, tag key, tag value, owning service or primary Region

1

Secret name	Description	Last retrieved (UTC)	Created on (UTC)
rds!db-d5f3b290-48e8-4b5f-ad2c-9115517f9565	The secret associated with the primary RDS DB instance: arn:aws:rds:ap-southeast-2:721274598579:db:stocklane-dev	29 October 2026	29 October 2026 at 19:14:06
stocklane-dev/platform	-	30 October 2026	29 October 2026 at 19:12:33
stocklane-dev/vision/database	-	30 October 2026	29 October 2026 at 19:12:31
stocklane-dev/workshop/database	-	30 October 2026	29 October 2026 at 19:12:29
stocklane-dev/inventory/database	-	30 October 2026	29 October 2026 at 19:12:27
stocklane-dev/grafana/admin	-	30 October 2026	29 October 2026 at 19:12:26

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KMS-backed database configuration

The PostgreSQL RDS configuration shows encrypted storage associated with a customer-managed KMS key.

Aurora and RDS

Dashboard
Databases
Query editor
CloudWatch Database Insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Capacity reservations New
Proxies

Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations

Events
Event subscriptions

Recommendations 0
Certificate update

stocklane-dev

Modify | Actions

Summary

DB identifier	Status	Role	Engine	Recommendations
stocklane-dev	Available	Instance	PostgreSQL	
CPU 3.47%	**Class** db.t4g.micro	**Current activity** 22 Connections	**Region & AZ** ap-southeast-2b	

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations

Instance

Configuration

DB instance ID
stocklane-dev

Engine version
17.9

RDS Extended Support
Enabled

DB name
stocklane

License model
Postgresql License

Option groups
default:postgres-17 In sync

Amazon Resource Name (ARN)
arn:aws:rds:ap-southeast-2:721274598579:db:stocklane-dev

Instance class

Instance class
db.t4g.micro

vCPU
2

RAM
1 GB

Availability

Master username
stocklane_admin

IAM DB authentication
Not enabled

Multi-AZ
No

Secondary Zone

Primary storage

Encryption
Enabled

AWS KMS key
stocklane-dev

Storage type
General Purpose SSD (gp2)

Storage
20 GiB

Provisioned IOPS
-

Storage throughput
-

Storage autoscaling
Disabled

Monitoring

Monitoring type
Database Insights - Standard

Database counter metrics
Disabled

Enhanced Monitoring
Disabled

Logs published to Cloudwatch
PostgreSQL log
Upgrade log

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H • IAM and trust

IAM roles and trust relationships give GitHub Actions, EKS and AWS services scoped access.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

GitHub trust roles

The filtered role list shows separate platform, release and training trust roles.

Identity and Access Management (IAM)
Roles
Policies
IAM users
Root access management

Roles (41) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Delete | Create role

git | 3 matches | 1

Role name	Trusted entities	Last activity
stocklane-dev-github-platform	Account: 721274598579	1 hour ago
stocklane-dev-github-release	Account: 721274598579	1 hour ago
stocklane-dev-github-training	Account: 721274598579	12 hours ago

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I • AI and ML

SageMaker executes model-processing jobs, while Textract and Bedrock support label understanding and reviewable suggestions.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

SageMaker processing jobs

Completed forecast and photo processing jobs provide execution evidence for the two model paths.

customization
Training & tuning jobs
JumpStart model hub New
HyperPod clusters
▶ Training plans
▼ Deployments & inference
Deployable models
Endpoints
Endpoint configurations
Batch transform jobs
Shadow tests
Compilation jobs
Inference Recommender
HyperPod inference
▼ Model governance
Model dashboard
Model cards
▶ AWS Marketplace resources
▼ Data preparation
Processing jobs
▶ Ground Truth
▶ Augmented AI

Tutorials
Documentation

Processing jobs

Actions ▾ | Create processing job

Find processing jobs

1

Name	ARN	Creation time	Duration	Status
stocklane-dev-public-forecast-20261029-122238-d5b0	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-forecast-20261029-122238-d5b0	10/29/2026, 5:52:38 PM	2 minutes	Completed
stocklane-dev-public-photo-20261029-122002-5bd1	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-photo-20261029-122002-5bd1	10/29/2026, 5:50:03 PM	2 minutes	Completed
stocklane-dev-public-forecast-20261029-101055-50fe	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-forecast-20261029-101055-50fe	10/29/2026, 3:40:55 PM	2 minutes	Completed
stocklane-dev-public-photo-20261029-100820-d25e	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-photo-20261029-100820-d25e	10/29/2026, 3:38:21 PM	2 minutes	Completed
stocklane-dev-public-forecast-20261029-092802-5211	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-forecast-20261029-092802-5211	10/29/2026, 2:58:03 PM	2 minutes	Completed
stocklane-dev-public-photo-20261029-092525-b3d7	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-dev-public-photo-20261029-092525-b3d7	10/29/2026, 2:55:26 PM	2 minutes	Completed
stocklane-processing-test-1793189671	arn:aws:sagemaker:ap-southeast-2:721274598579:processing-job/stocklane-processing-test-1793189671	10/28/2026, 5:44:32 PM	a minute	Completed

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J • Observability and alerts

CloudWatch, EventBridge, Lambda and SNS turn application and queue signals into actionable notifications.

- The screenshots in this group correspond to one part of the AWS infrastructure diagram.

CloudWatch overview

CloudWatch shows application and SQS alarm signals for the platform.

CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms 0 5 0
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Container Insights
Database Insights
Lambda Insights
SageMaker Insights New
EC2 Resource Health
All Insights
Logs
Log Management New
Log Analytics New
Log Anomalies
Omni New
Observability Workshops

Overview info

1h 3h 12h 1d 1w Custom | UTC timezone

Overview | Filter by resource group | info | Actions

CloudWatch Omni - Observability for the AI Era New

Unified observability experience for your applications and AI workloads, delivered outside of the AWS console with native AI-powered intelligence. Learn more

Get started with Omni

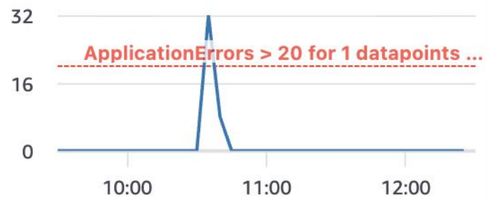
Alarms by AWS service info

Services	In alarm 0	Insufficient data 0	OK 5
Stocklane/stocklan…			(1)
Simple Queue Service			(4)

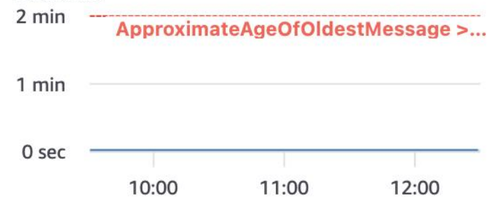
Recent alarms info

View recent alarms dashboard

stocklane-dev-applicatio…



stocklane-dev-stock-bac…



Application Monitoring info

Real User Monitoring info

Top 5 | View all

View in RUM

Get started by creating your first application monitor.

Create application monitor

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EventBridge RCA rule

The event bus routes alarm events to the Stocklane root-cause workflow.

Amazon EventBridge

Dashboard
Event buses New

Pipes
Pipes

Scheduler
Schedules
Schedule groups
Scheduled rules (legacy) Updated

Integration
Partner event sources
API destinations
Connections

Schema registry
Schemas

Classic event routing
Global endpoints

Developer resources
Learn
Sandbox
Quick starts

Documentation
Free virtual workshops New

Event bus details

Event bus name
default

Description
-

Event bus ARN
arn:aws:events:ap-southeast-2:721274598579:event-bus/default

Schema discovery
Not Initiated

Discoverer ID
No discoverer

Discoverer ARN
No discoverer

Created on
-

Last modified
-

Rules | Encryption | Permissions | Monitoring | Logs | Archives and replays | Tags

Event pattern rules (2)

Delete | Enable | Edit | CloudFormation Template | Create rule

We have moved Scheduled rules in your account to 'Scheduler' section, where you can view, create and edit a scheduled rule. Go to Scheduled rules

Find rules | Any status | 1

Name	Status	Type	Event bus	ARN	Description
AutoScalingManagedRule	Enabled	Managed	default	arn:aws:events:ap-southeast-2:721274598579:rule/AutoScalingManagedRule	This rule is used to route Instance notifications to EC2 Auto Scaling
stocklane-dev-alarm-rca	Enabled	Standard	default	arn:aws:events:ap-southeast-2:721274598579:rule/stocklane-dev-alarm-rca	Project alarms entering ALARM start a rule-based root-cause report

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Lambda RCA function

The Lambda function receives alarm context and produces root-cause information.

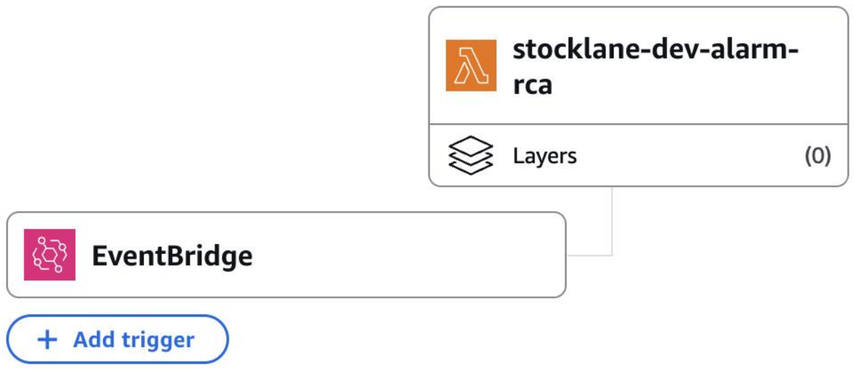
stocklane-dev-alarm-rca

Throttle | Copy ARN | Actions ▼

Function overview Info

Export to Infrastructure Composer | Download ▼

Diagram | Template



Function ARN
arn:aws:lambda:ap-southeast-2:721274598579:function:stocklane-dev-alarm-rca

Description
-

Last modified
18 hours ago

Code | Test | Monitor | Configuration | Aliases | Versions

Executing function: succeeded (logs)
▶ Details

Test event Info

CloudWatch Logs Live Tail | Save | Test

To invoke your function without saving an event, configure the JSON event, then choose Test.

Test event action

(•) Create new event ( ) Edit saved event

Invocation type

(•) Synchronous
Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

Info | Tutorials

Choose a tutorial

Learn how to implement common use cases in AWS Lambda.

Create a simple web app

In this tutorial you will learn how to:

- Build a simple web app using a Lambda Web Function that returns an HTML webpage
- Invoke your Web Function through its endpoint URL

Start tutorial

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SNS RCA notification

The resulting notification communicates the alarm and recommended investigation actions.

2 of 6,723

[Stocklane RCA] stocklane-dev-application-errors

Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com>

4:08 PM (3 hours ago)

to me

Alarm: stocklane-dev-application-errors
Reason: Threshold Crossed: 1 datapoint [28.0 (30/10/26 10:33:00)] was greater than the threshold (20.0).
Window: 2026-10-30 10:08 to 10:38 UTC

LIKELY CAUSE (rule-based, verify before acting)
Most errors in the window came from kube-state-metrics in monitoring (189 error lines). 10 AWS change(s) happened shortly before, starting with Mfa_POST (cognito-idp.amazonaws.com).

RECOMMENDED ACTIONS
1. Check kube-state-metrics pod status and restarts
2. Read the sample error lines below
3. Review the recent changes listed below; roll back if one matches the start of the alarm

ALARM METRIC (latest values)
10:08 0
10:13 0
10:18 0
10:23 0
10:28 0
10:33 28

TOP ERROR SOURCES
monitoring/kube-state-metrics: 189
kube-system/csi-snapshotter: 184
stocklane-dev/inventory: 41
monitoring/grafana: 41
argocd/argocd-application-controller: 9

CONCLUSION

Conclusion

Stocklane combines a practical workshop workflow with a production-minded AWS platform. React provides the operator experience; Spring Boot and FastAPI provide focused microservices; GitHub Actions and Terraform automate delivery; Argo CD reconciles EKS; MLOps and LLMOps make AI useful but reviewable; and AWS services provide the secure, observable foundation.

- Business flow: scan or search → review → approve → record.
- Engineering flow: commit → validate → provision → release → sync → observe.
- The architecture and screenshots now follow one consistent service-based story.